Model Quantization

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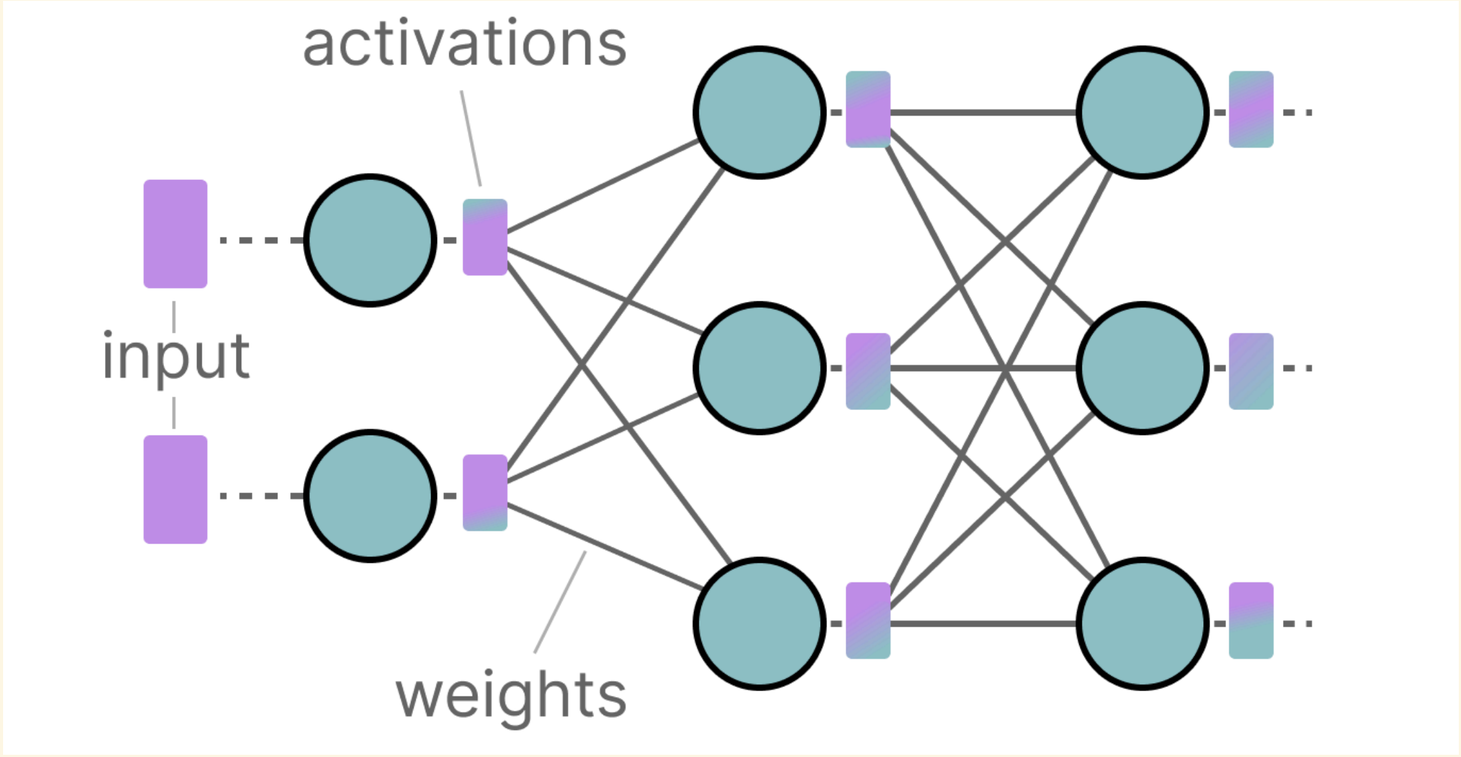
Why LLMs need so much memory

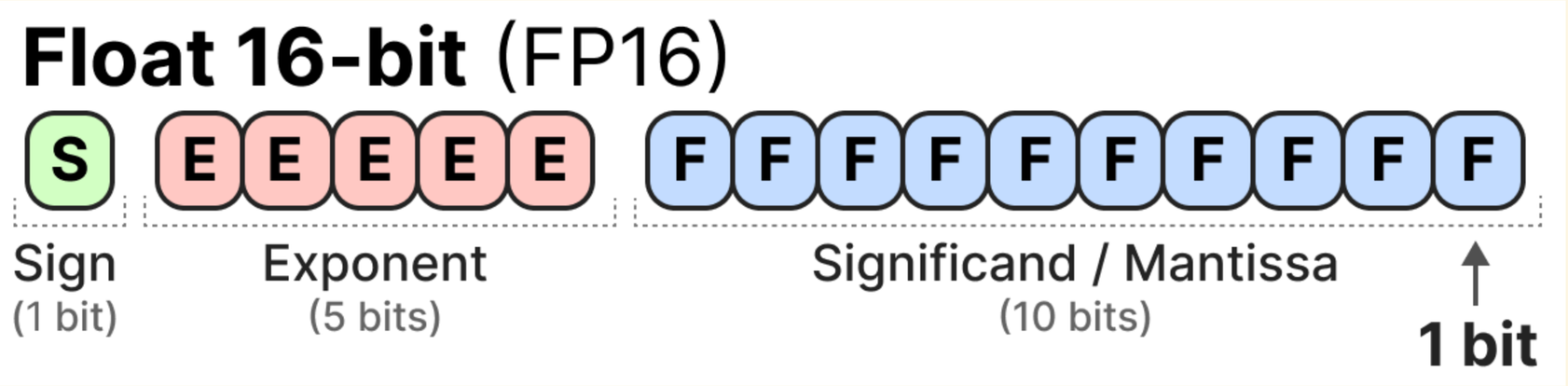
Every layer of a neural network stores two kinds of numbers: **weights** (the learned parameters, fixed after training) and **activations** (the values flowing through the network on a given input, different every forward pass). Both are stored as floating-point numbers, and both take memory:



A model with billions of weights means billions of floating-point numbers resident in memory at once — **how many bits each number takes** is therefore the single biggest lever on how much memory the model needs.

Floating point: sign, exponent, mantissa

A floating-point format splits its bits into three fields (the IEEE-754 standard):

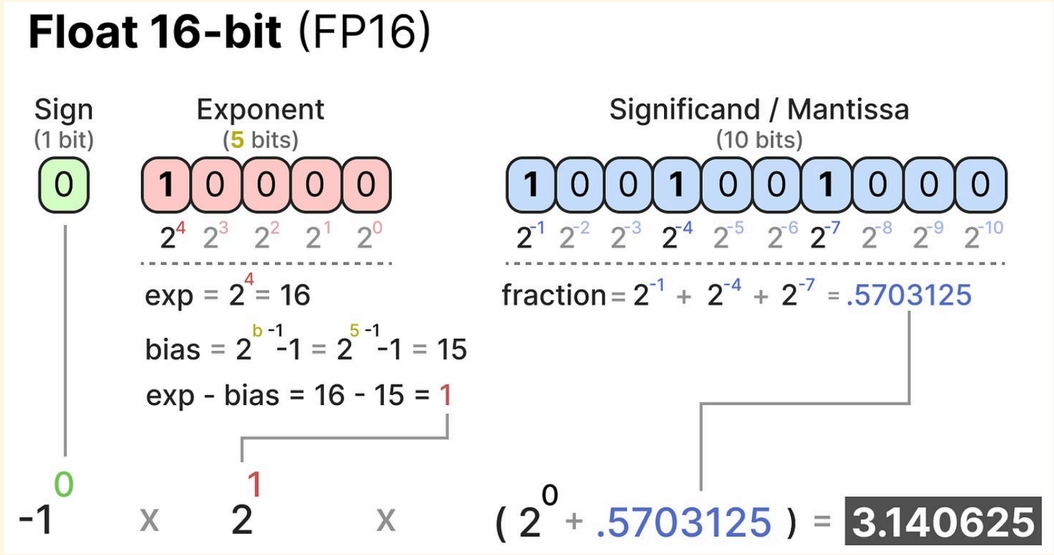


$$\text{value} = (-1)^{S} \times 2^{\,E-\text{bias}} \times \left(1 + \sum_{k=1}^{m} F_k \, 2^{-k}\right)$$

S — the sign bit; E — the exponent field read as an unsigned integer; bias — a fixed offset so the exponent can represent negative powers of 2, $\text{bias} = 2^{e-1} - 1$ for e exponent bits; F_k — the k-th mantissa bit (1 or 0), m mantissa bits; the leading $1 + (\cdot)$ is an implicit bit — never stored, assumed for every normal (non-zero, non-subnormal) number.

Decoding a float: a worked example

Take the FP16 bit pattern below and recover the number it encodes, one field at a time:



STEP	COMPUTATION	RESULT
sign	$S = 0 \Rightarrow (-1)^0$	$+1$
exponent	$E = 10000_2 = 16$, bias $= 2^4 - 1 = 15$	$16 - 15 = 1$
mantissa	bits $1001001000 \Rightarrow 2^{-1}+2^{-4}+2^{-7}$	0.5703125
value	$(+1) \times 2^1 \times (1 + 0.5703125)$	3.140625

> **THIS IS π, TO THE PRECISION FP16 ALLOWS**
>
> The true value $\pi = 3.14159265\ldots$ gets rounded to the nearest representable FP16 value, 3.140625 — an error of about 0.00097, purely from having only 10 mantissa bits to work with.

Where a format's min and max come from

The largest and smallest representable magnitudes follow directly from how many bits go to the exponent and mantissa. The all-ones exponent is reserved for $\pm\infty$/NaN, so the largest *usable* stored exponent is $2^e - 2$; the smallest is 1 (stored exponent 0 is reserved for zero/subnormals):

MAX MAGNITUDE

$$(2 - 2^{-m}) \times 2^{(2^e - 2) - \text{bias}}$$

All mantissa bits set (closest to 2), largest usable exponent.

MIN NORMAL MAGNITUDE

$$1.0 \times 2^{1 - \text{bias}}$$

All mantissa bits zero, smallest usable exponent.

FORMAT	SIGN+EXP+MANTISSA	BIAS	MAX	MIN (NORMAL)
FP32	1 + 8 + 23	127	3.4028×10^{38}	1.1755×10^{-38}
FP16	1 + 5 + 10	15	6.5504×10^{4}	6.1035×10^{-5}
BF16	1 + 8 + 7	127	3.3895×10^{38}	1.1755×10^{-38}

RANGE IS EXPONENT-DOMINATED, NOT EXPONENT-ONLY

Min normal has zero mantissa-dependence: it's always $1.0 \times 2^{1-\text{bias}}$, mantissa forced to all-zero. **Max** does depend on the mantissa a little, through the $(2 - 2^{-m})$ term above — that's exactly why the table shows BF16's max (3.3895×10^{38}) isn't bit-identical to FP32's (3.4028×10^{38}) despite sharing all 8 exponent bits. But $(2 - 2^{-m})$ only ever ranges between 1 and 2 — at most a $2\times$ difference — dwarfed by the exponent's contribution: 2^{127} vs 2^{15} is a factor of 2^{112}. So the mantissa can nudge the max by less than 1 bit's worth of magnitude, while the exponent moves it by 112 bits' worth — which is the precise sense in which range is set by the exponent, and FP16's 3-bits-narrower exponent (not its mantissa) is why its max is 65504, not 10^{38}.

The max, bit by bit

Concretely, which bit pattern *is* FP16's max? All mantissa bits set, largest usable exponent, exactly as the previous slide's formula describes:

STEP	COMPUTATION	RESULT
sign	$S = 0 \Rightarrow (-1)^0$	$+1$
exponent	$E = 11110_2 = 30$ (all-1s, 11111, is reserved for ∞/NaN — so the largest *usable* pattern is one less)	$30 - 15 = 15$
mantissa	all 10 bits set: $1111111111 \Rightarrow 1023/1024$	0.9990234375
value	$(+1) \times 2^{15} \times (1 + 0.9990234375)$	65504

This is exactly the 6.5504×10^4 from the previous slide's table — not a separately quoted constant, but this specific bit pattern, decoded with the same formula used for every other value. One exponent step higher would need the reserved all-1s pattern, which is why 65504 is the ceiling.

Subnormal numbers: the other reserved exponent

Just as all-1s exponent bits are reserved (for $\pm\infty$/NaN), all-**0s** exponent bits are reserved too — for numbers *smaller* than any normal number can represent. When $E = 0$, the format switches formulas: the implicit leading 1 disappears, and the exponent freezes at the smallest normal's value instead of continuing to shrink:

NORMAL ($E \neq 0$)

$$(-1)^S \times 2^{E-\text{bias}} \times (1 + F)$$

SUBNORMAL ($E = 0$, $F \neq 0$)

$$(-1)^S \times 2^{1-\text{bias}} \times F$$

F — the mantissa read as a plain fraction ($F = \sum F_k 2^{-k}$, no added 1). Dropping the implicit 1 is what lets these values shrink past the normal floor — each step down the mantissa halves the value instead of barely denting it, so precision degrades gracefully instead of the format jumping straight from its smallest normal number to exactly 0 (**gradual underflow**). If $E = 0$ and $F = 0$ too, the value is exactly 0.

BIT PATTERN (S E F)	COMPUTATION	VALUE
0 00000 1000000000	$2^{1-15} \times (512/1024)$	3.0518×10^{-5}
0 00000 0000000001	$2^{1-15} \times (1/1024)$	5.9605×10^{-8} (smallest subnormal)
0 00001 0000000000	$2^{1-15} \times (1 + 0)$	6.1035×10^{-5} (smallest *normal*)

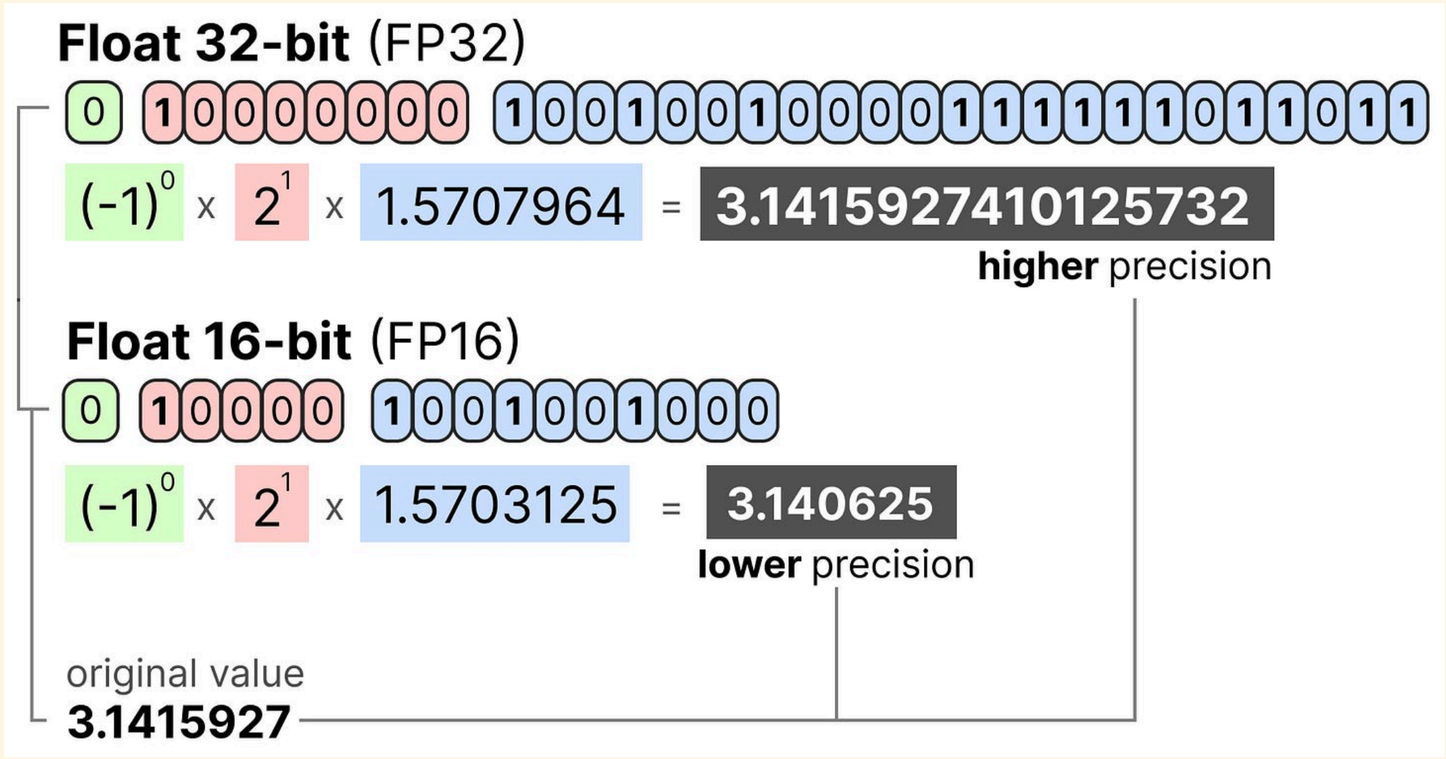
So FP16's true floor isn't 6.1×10^{-5} — it's 5.96×10^{-8}, once subnormals are counted. A value can still be too small even for that: 1.8×10^{-42}, coming up shortly, is far below even this subnormal floor, which is why it underflows all the way to 0.0 rather than landing on a subnormal.

SUBNORMALS ARE NOT "EVERYTHING BELOW 1"

Almost all of $(0, 1)$ — 0.5, 0.1, 0.001, down to about 6.1×10^{-5} — is still **normal**, just with a negative exponent ($0.1 = 2^{-4} \times 1.6$, e.g.). Subnormals only take over in the last, extremely thin sliver right before zero.

Same value, different precision

π encoded in FP32 and FP16 side by side — more mantissa bits means the rounded value sits closer to the true one:



Both use the same $2^{E-\text{bias}}$ term (exponent $= 1$ either way); only the fraction changes — FP32's 23 mantissa bits place π within 10^{-8}, FP16's 10 bits within 10^{-3}.

When a value falls outside the range

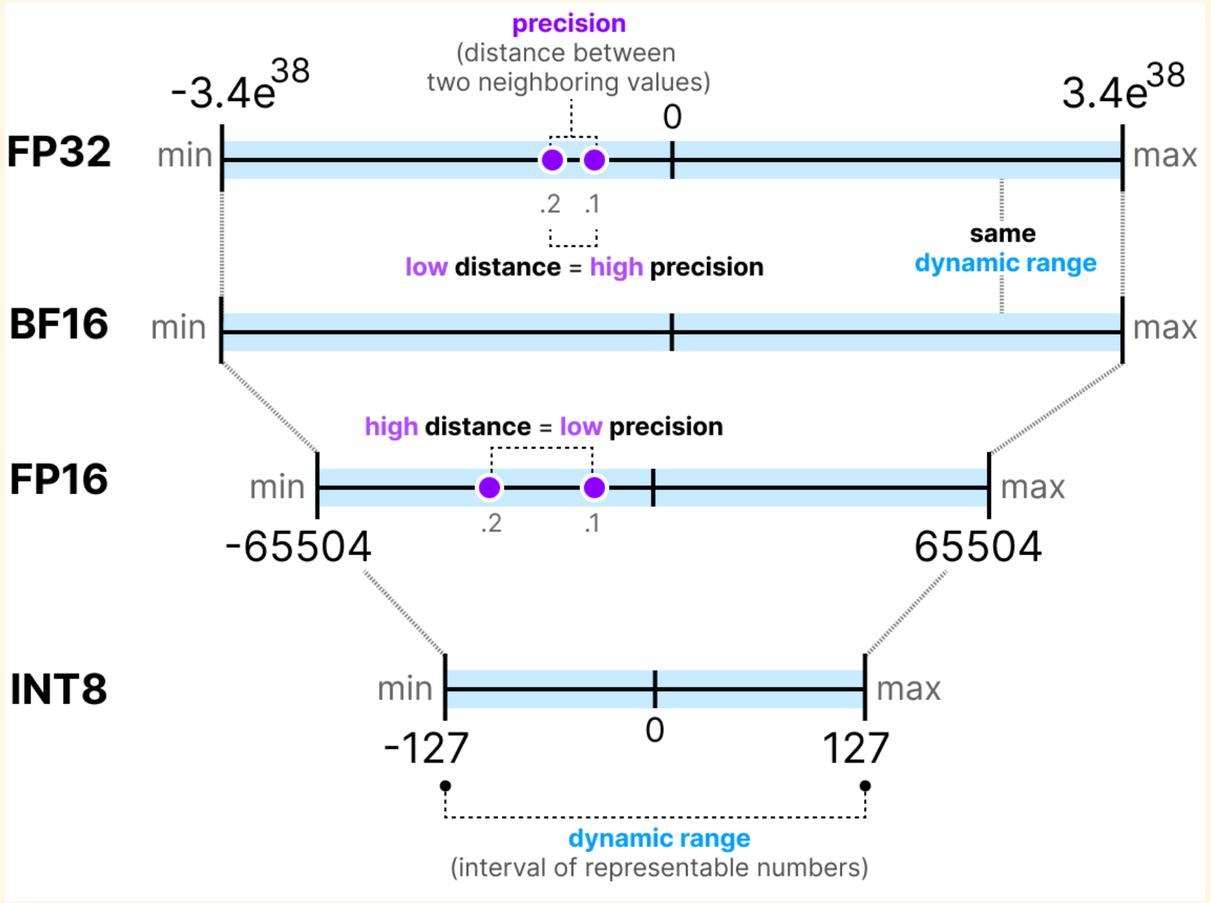
Two numbers, cast to progressively narrower formats — one is an ordinary magnitude, the other deliberately tiny:

Original	75505.0	1.8e-42
64-bits	75505.0	1.8e-42
32-bits	75505.0	1.80066e-42
16-bits	inf	0.0

75505.0 is an ordinary magnitude — well inside FP32's range, and still fine at FP16's own range… except FP16's max is 65504 (a few slides back), and $75505.0 > 65504$. It **overflows**: there's no representable FP16 value large enough, so it rounds to $+\infty$. 1.8×10^{-42} fails at the *opposite* end — it's below FP16's minimum normal magnitude (6.1×10^{-5}) and even below FP16's subnormal floor, so it **underflows to exactly 0.0**. Same failure, two different directions: one value is too big for the range, the other too small.

Dynamic range vs. precision

Two separate properties of a format, easy to conflate:



Dynamic range — the interval of representable magnitudes (dominated by exponent bits; the earlier min/max slide has the small mantissa contribution too).
Precision — the gap between two neighboring representable values (set by mantissa bits, for floats). BF16 trades FP16's precision for FP32's exponent range; INT8 has neither an exponent nor a mantissa — its whole range is fixed and evenly spaced.

The memory cost of a data type

Every parameter costs its bit-width, no more and no less:

$$\text{memory} = \frac{\text{nr\_bits}}{8} \times \text{nr\_params}$$

nr_bits — bits per stored parameter (32 for FP32, 16 for FP16/BF16, 8 for INT8, ...); nr_params — the model's total parameter count; dividing by 8 converts bits to bytes.

A 70-billion-parameter model, in memory

Applying the formula from the previous slide to a 70B-parameter model at three precisions:

$$\textbf{64-bits} = \frac{64}{8} \times 70B \approx \textbf{560}\ GB$$

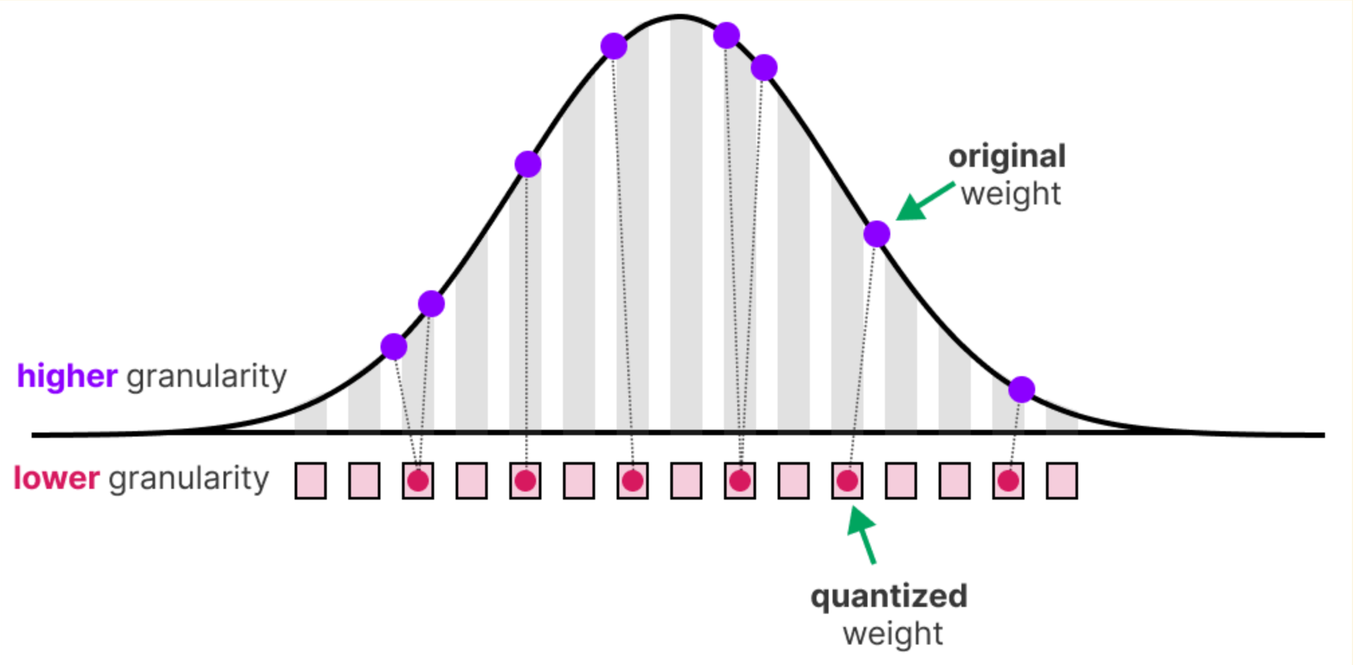
$$\textbf{32-bits} = \frac{32}{8} \times 70B \approx \textbf{280}\ GB$$

$$\textbf{16-bits} = \frac{16}{8} \times 70B \approx \textbf{140}\ GB$$

Halving the bit-width exactly halves the memory — going from FP32 to FP16/BF16 alone turns a model that needs **280GB** (more than any single consumer GPU) into one that needs **140GB**. Quantization pushes this idea further, down to 8 or even 4 bits per weight.

What is quantization?

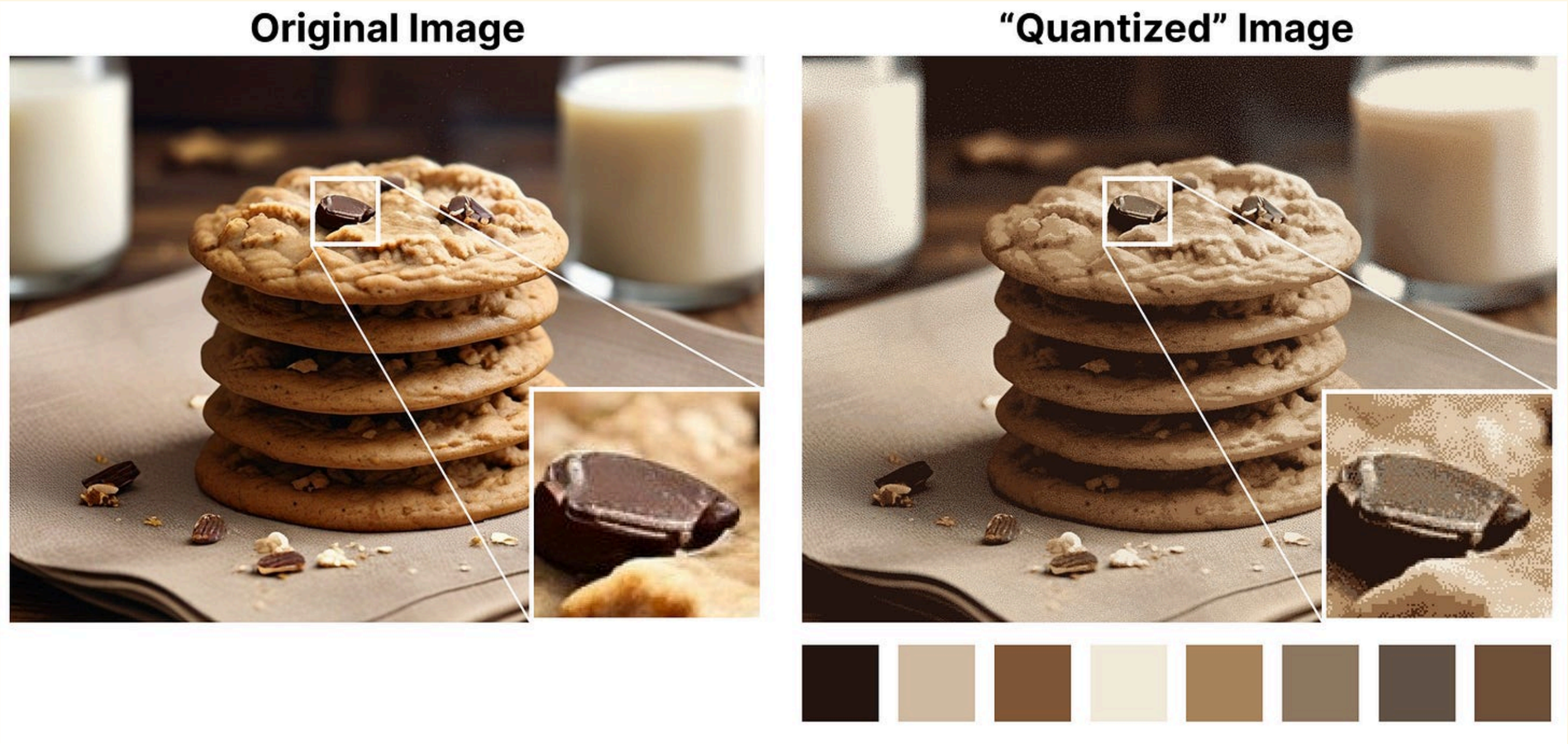
Quantization maps each original value onto the **nearest point of a coarser, lower-granularity grid** — trading precision for a far smaller representation:



The gap between the original weight and its quantized snap-point is the **quantization error** — small enough, in aggregate across billions of weights, that the model's behavior barely changes, even though every individual number moved.

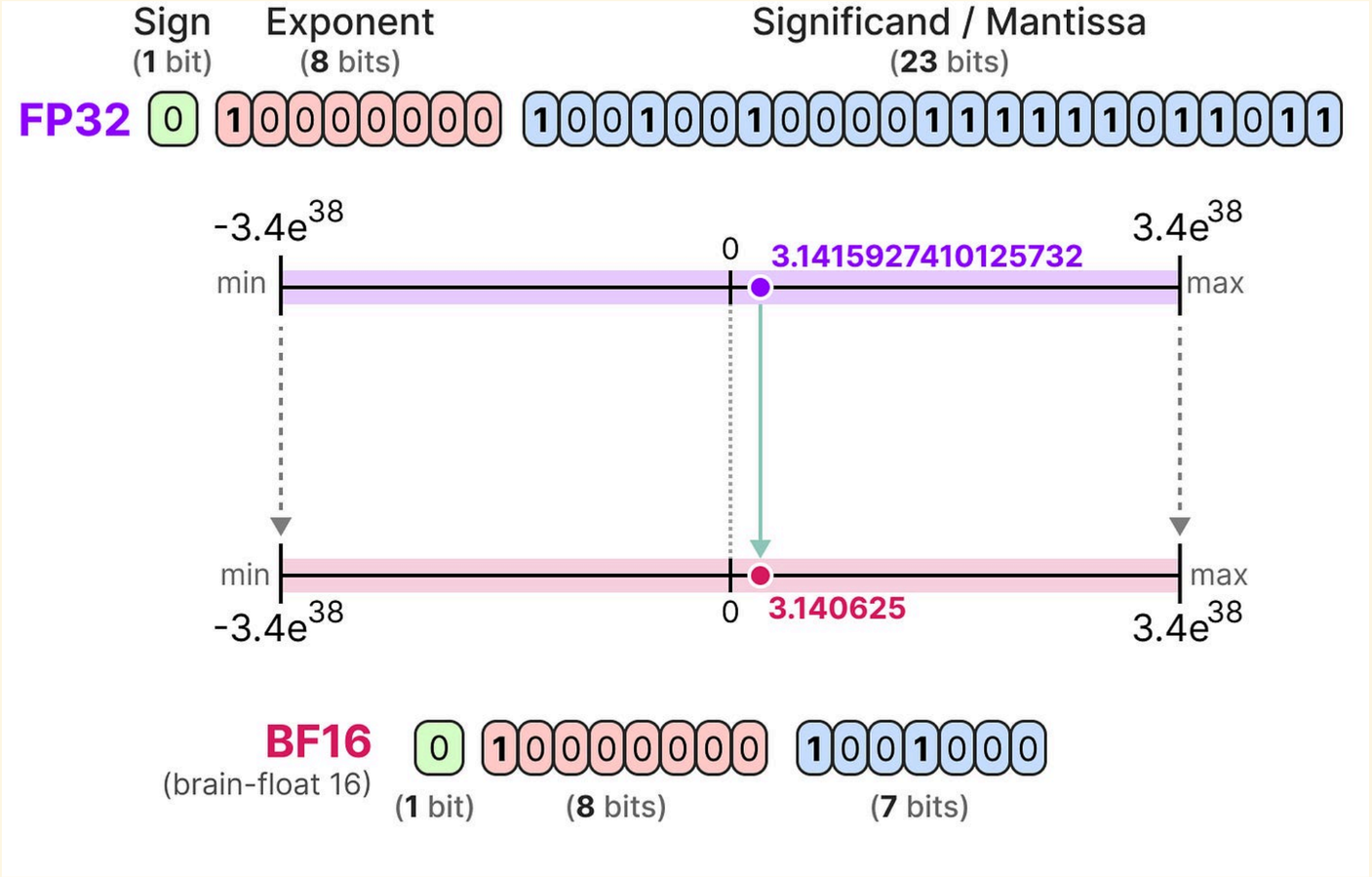
The same idea, in an image

Reducing a photo from millions of colors to an 8-color palette is the same operation — every pixel snaps to its nearest available color:



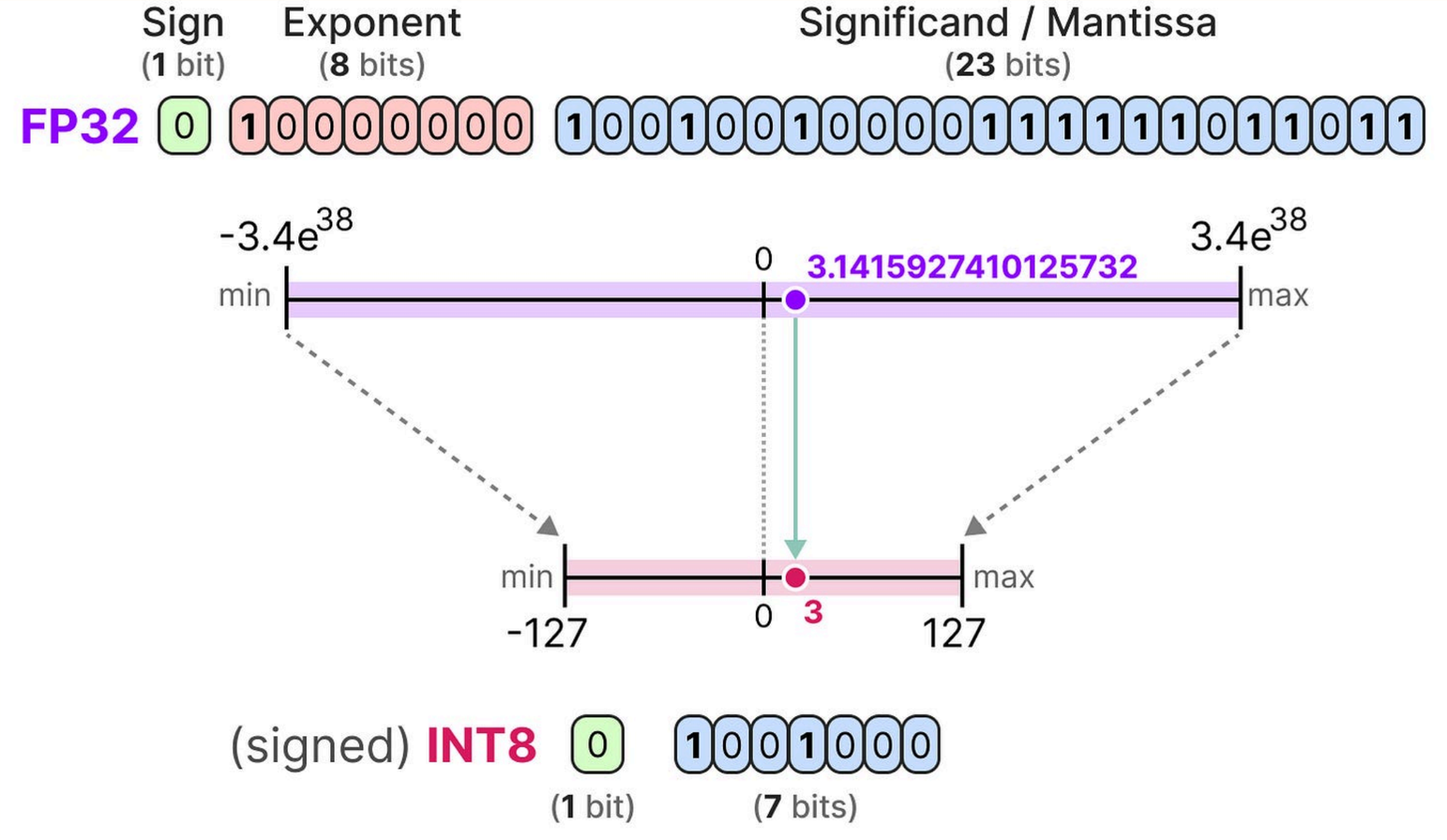
Structure survives (the cookies are still clearly cookies); fine gradation is what's lost — exactly the trade-off model weight quantization makes.

BF16: FP32's range, less precision



BF16 keeps FP32's 8 exponent bits (same dynamic range, same min/max) but cuts the mantissa to 7 bits, so it's cheaper to store than FP32 while far less likely to overflow/underflow than FP16 — the reason it's the default training dtype for most modern LLMs.

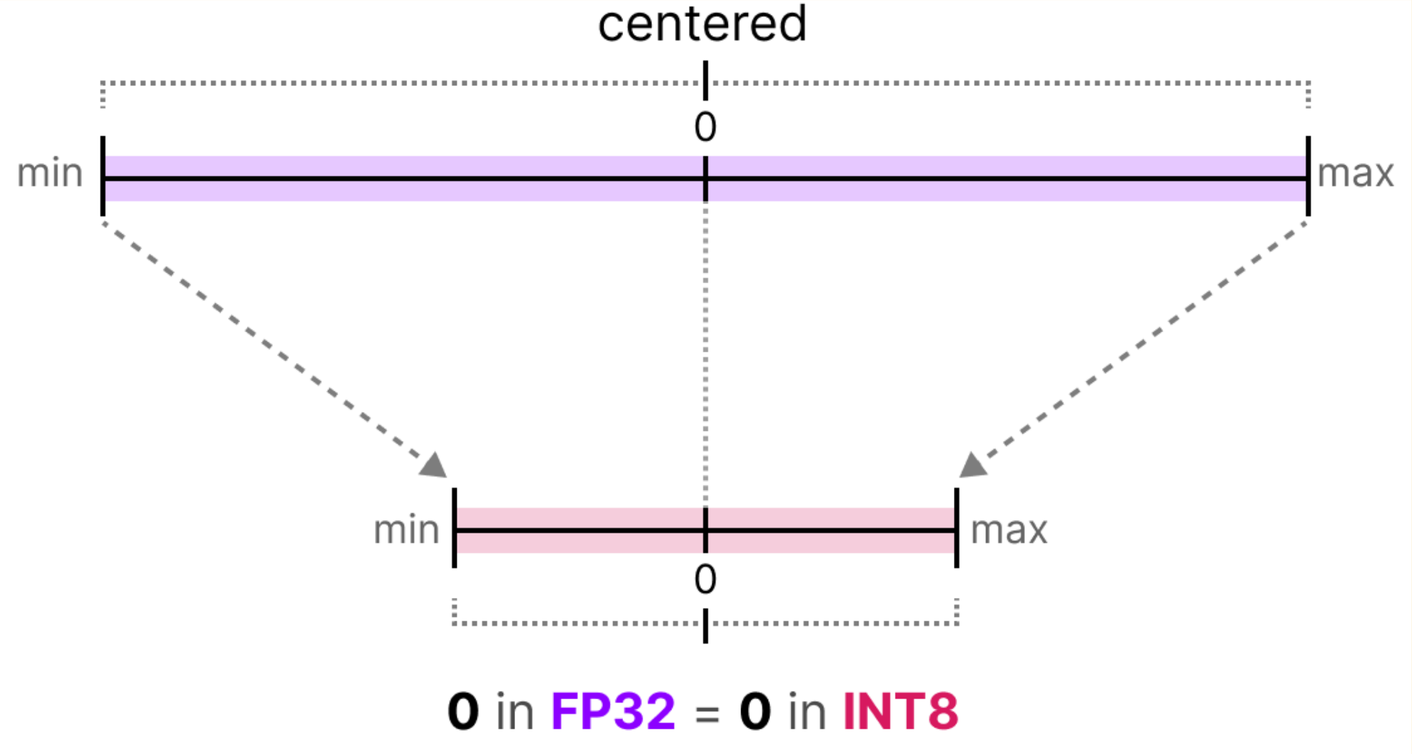
INT8: a fixed integer grid



INT8 has no exponent or mantissa — just a sign and 7 bits of magnitude, giving exactly 256 evenly-spaced integers, -127 to 127 (signed). There's no built-in notion of where the decimal point goes; **quantization is exactly the scheme that decides that**, covered next.

Symmetric quantization

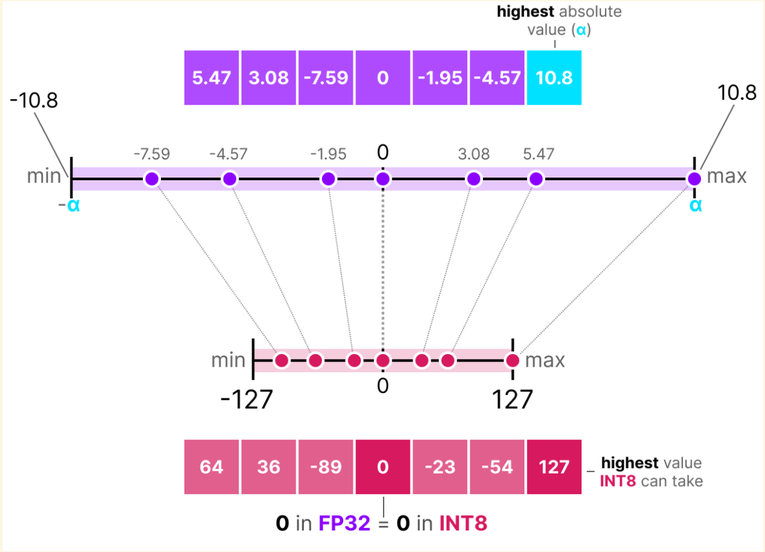
Map the range $[-\alpha, \alpha]$ — centered on the original data's largest magnitude — onto the integer range $[-(2^{b-1}-1),\ 2^{b-1}-1]$, so that 0 always maps to 0:



$$s = \frac{2^{b-1}-1}{\alpha}, \qquad x_{\text{quantized}} = \text{round}(s \cdot x)$$

α — the highest absolute value in the tensor being quantized; b — the number of **bits** per stored value, not bytes ($b = 8$ bits $\Rightarrow 2^{b-1}-1 = 127$, the INT8 range from Part 1); s — the scale factor, real-valued; x — an original FP32 value; $x_{\text{quantized}}$ — the stored integer; $\text{round}(\cdot)$ — round to the nearest integer, same as everyday rounding (e.g. $\text{round}(64.33) = 64$, $\text{round}(-89.26) = -89$) — this is the only step that actually loses information; everything else here is exact arithmetic.

Symmetric quantization: a worked example



$x = [5.47,\ 3.08,\ -7.59,\ 0,\ -1.95,\ -4.57,\ 10.8]$, so $\alpha = \max|x| = 10.8$ and $s = 127/10.8 = 11.76$:

X	$S \cdot X$	$X_{\text{QUANTIZED}} = \text{ROUND}(S \cdot X)$
5.47	64.33	64
3.08	36.22	36
−7.59	−89.26	−89
0	0	0
−1.95	−22.93	−23
−4.57	−53.74	−54
10.8	127.0	127

Symmetric quantization: dequantizing back

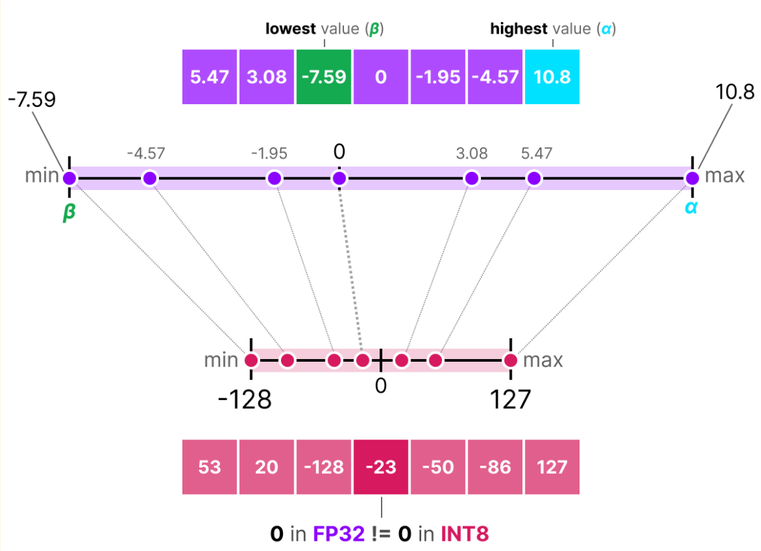
$$x_{\text{dequantized}} = \frac{x_{\text{quantized}}}{s}, \qquad \text{error} = \left| x - x_{\text{dequantized}} \right|$$

x	$x_{\text{quantized}}$	$x_{\text{dequantized}}$	error
5.47	64	5.44	0.03
3.08	36	3.06	0.02
−7.59	−89	−7.57	0.02
0	0	0.00	0.00
−1.95	−23	−1.96	0.01
−4.57	−54	−4.59	0.02
10.8	127	10.80	0.00

Dividing each stored integer back by $s = 11.76$ recovers values close to, but not exactly, the originals. The error is at most about 0.03 here, and exactly 0 wherever a value happened to round perfectly — like 10.8, which mapped straight onto the grid's own edge ($x_{\text{quantized}} = 127$) with no rounding at all.

Asymmetric quantization

Symmetric quantization wastes range when data isn't centered on zero. Asymmetric quantization maps the data's *actual* min/max, $[\beta, \alpha]$, onto the full integer range and shifts the origin with a **zero-point** z:



$$s = \frac{2^b - 1}{\alpha - \beta}, \qquad z = \text{round}(-s \cdot \beta) - 2^{b-1}, \qquad x_{\text{quantized}} = \text{round}(s \cdot x + z)$$

β — the lowest value in the tensor; α — the highest value; z — the integer zero-point, the stored code that FP32's 0 maps to (generally $\neq 0$, unlike the symmetric case).

> WHERE Z COMES FROM: SCALE FIRST, THEN SLIDE
>
> s alone only fixes the interval's *width* to match the integer range's width — after scaling, β and α have the right spread but not necessarily the right position. z is the leftover **shift** needed to slide $s \cdot \beta$ exactly onto the integer floor -2^{b-1}: $z = -2^{b-1} - s \cdot \beta$. Setting $x = 0$ in $s \cdot x + z$ then gives z itself — the same shift that aligns the minimum also happens to be where zero lands, which is why it's called the zero-point.

Deriving z: two examples

Demand $x = \beta$ maps to the integer floor -2^{b-1}, before rounding: $s\beta + z = -2^{b-1} \Rightarrow z = -2^{b-1} - s\beta$. Solving using only β is enough — α lands on the ceiling automatically, since s already stretched the span to fit:

Mixed-sign data: $\beta = -2,\ \alpha = 6$

$$s = \frac{255}{6 - (-2)} = 31.875$$

$$z = \text{round}(31.875 \times 2) - 128 = -64$$

X	$SX + Z$	CODE
$\beta = -2$	-127.75	-128 (floor)
0	-64.00	$-64 = z$
$\alpha = 6$	127.25	127 (ceiling)

All-positive data: $\beta = 10,\ \alpha = 20$

$$s = \frac{255}{20 - 10} = 25.5$$

$$z = \text{round}(-25.5 \times 10) - 128 = -383$$

X	$SX + Z$	CODE
$\beta = 10$	-128.00	-128 (floor)
0	-383.00	never occurs
$\alpha = 20$	127.00	127 (ceiling)

> **Z DOESN'T HAVE TO BE A VALID CODE ITSELF**
>
> Here $z = -383$ is far outside $[-128, 127]$ — and that's fine. Real zero never actually occurs in this tensor (every value is ≥ 10), so nothing ever needs to be stored at code -383; z only has to make the formula land correctly for $x \in [\beta, \alpha]$, which it does exactly.

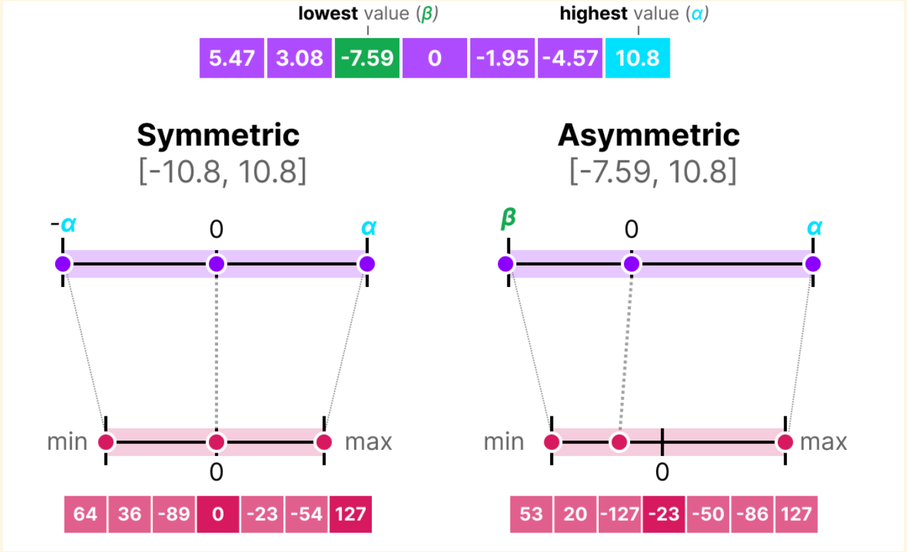
Asymmetric quantization: a worked example

Same vector as before: $\beta = \min(x) = -7.59$, $\alpha = \max(x) = 10.8$, so $s = 255/18.39 = 13.86$ and $z = \text{round}(13.86 \times 7.59) - 128 = 105 - 128 = -23$:

X	$S \cdot X + Z$	$X_{\text{QUANTIZED}}$	DEQUANTIZED $(X_{\text{QUANTIZED}} - Z)/S$
5.47	52.82	53	5.481
3.08	19.69	20	3.101
−7.59	−128.19	−128 (clamped)	−7.572
0	−23	−23	0.0
−1.95	−50.03	−50	−1.947
−4.57	−86.34	−86	−4.543
10.8	126.7	127	10.818

Note -7.59 needed -128.19, outside even the asymmetric grid's floor — it gets clamped to -128, the largest quantization error in this example.

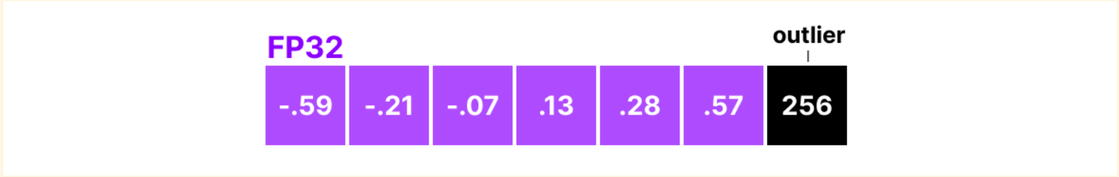
Symmetric vs. asymmetric, side by side

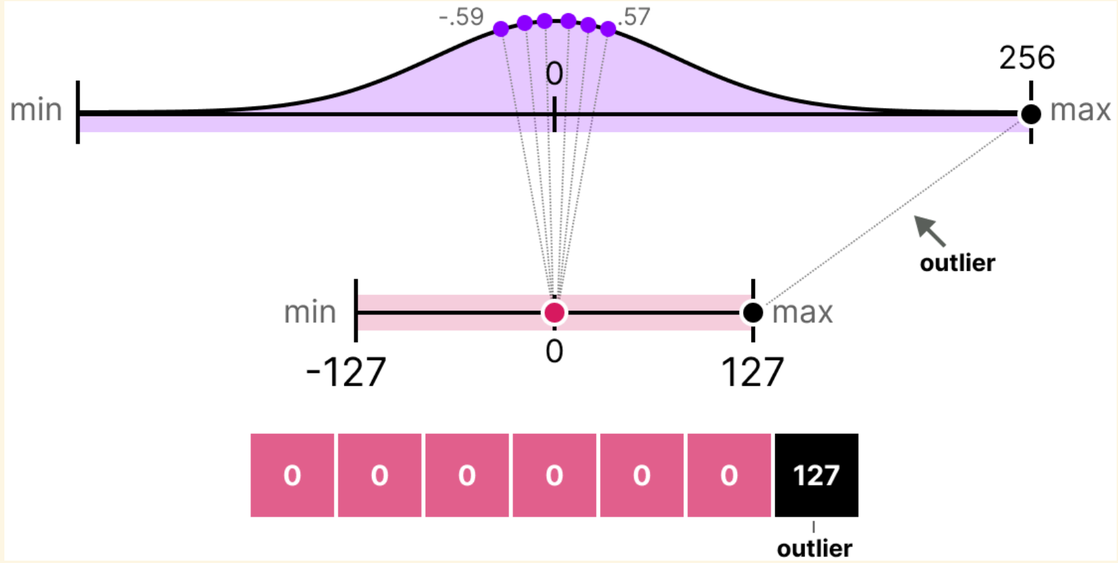


Asymmetric quantization uses the integer grid more fully whenever the data isn't centered on zero — at the cost of tracking one extra number, z, and a slightly more expensive dequantization step.

> WHY $[-127, 127]$ HERE BUT $[-128, 127]$ THERE
>
> INT8 spans $[-128, 127] - 256$ codes, asymmetric since two's-complement has no separate $+0/-0$. Symmetric quantization gives up code -128 so $\pm\alpha$ land on *equal-magnitude* codes (∓ 127) with 0 exactly centered — that mirror symmetry is the whole point. Asymmetric never needed $0 \mapsto 0$, so it keeps all 256 codes.

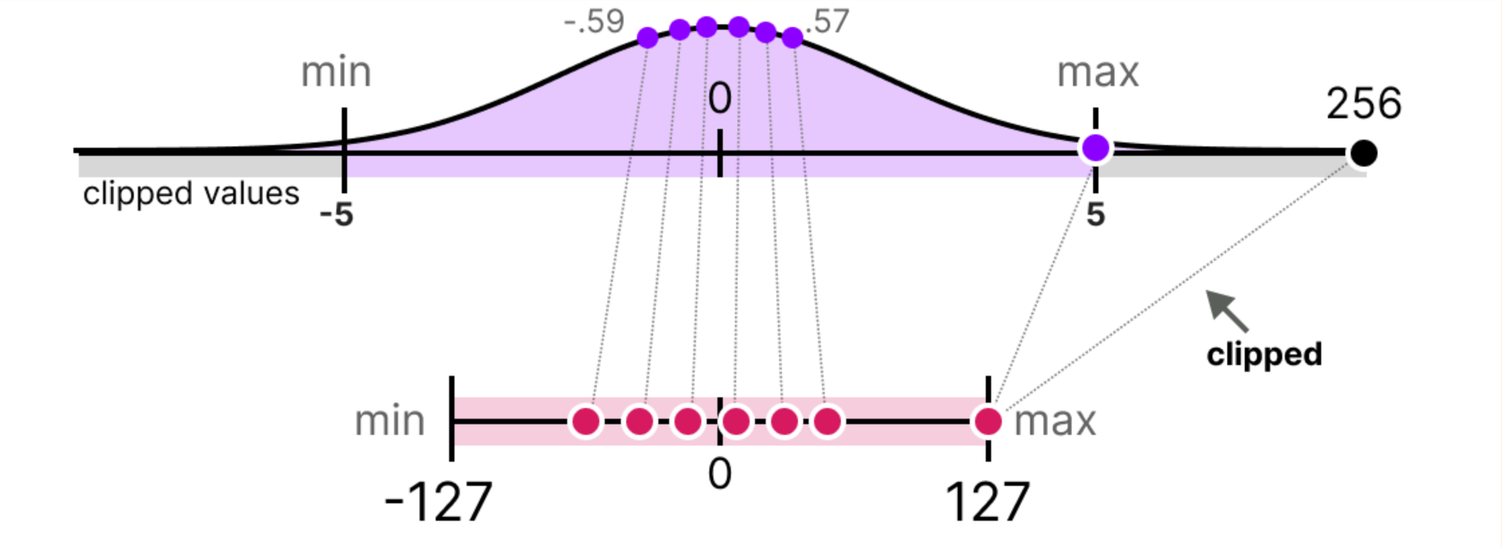
One outlier ruins the whole grid





$\alpha = \max|x|$ is set entirely by the single outlier (256), so $s = 127/256 \approx 0.496$ is tiny — every other value, all under 1.0 in magnitude, rounds straight down to **0**. A single extreme value can silently destroy the precision of every normal value sharing its scale factor.

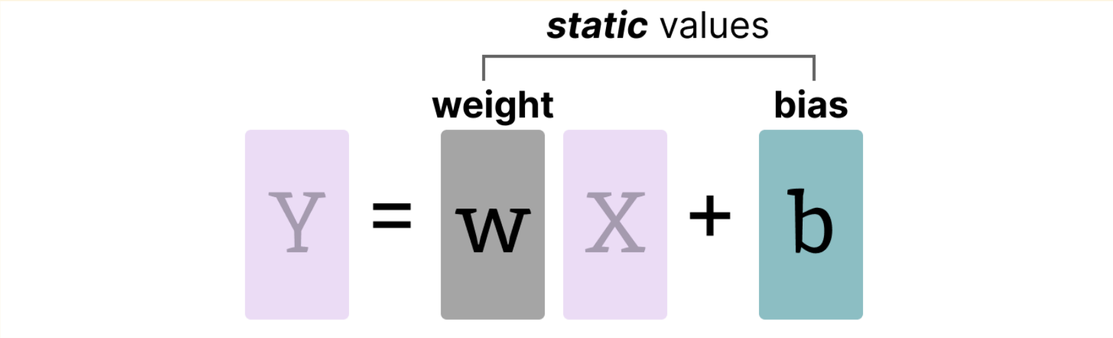
Clipping: trading outlier accuracy for the rest

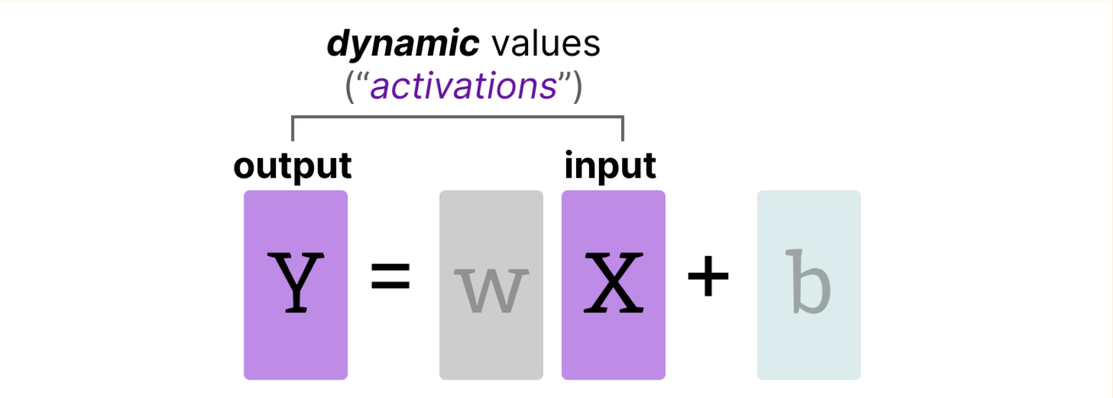


Clip the range to $[-5, 5]$ before computing α: the outlier gets truncated to 5 (and quantizes with large error), but $s = 127/5 = 25.4$ is now large enough that every normal value gets a distinct, meaningful integer code instead of collapsing to 0. Choosing *where* to clip is exactly what calibration (next) is for.

Calibration: what range do we clip to?

Every quantization scheme needs a $[\beta, \alpha]$ (or α) range before it can compute s — **calibration** is how that range gets chosen, and it looks different for the two kinds of values a network holds:

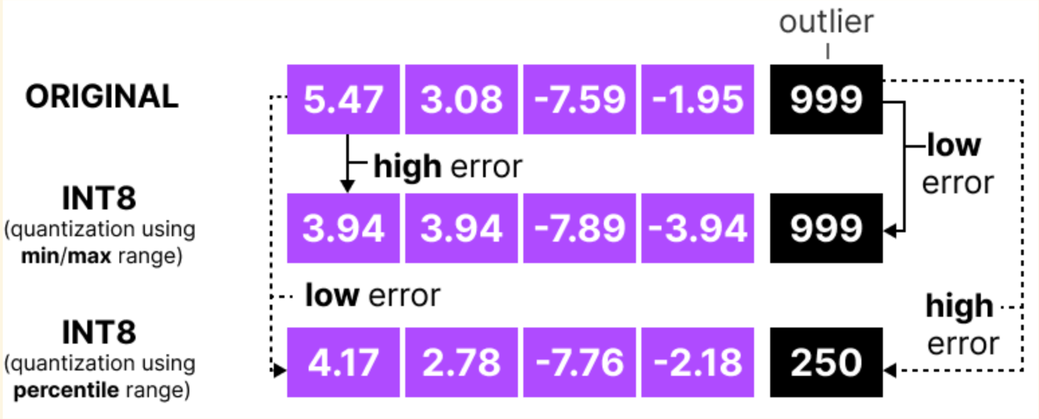




Weights are fixed after training — their range can be measured once, in advance. **Activations** depend on the current input, so their range shifts on every forward pass — calibrating them means deciding whether to measure that range on the fly or fix it ahead of time (Part 3).

Min/max isn't the only calibration choice

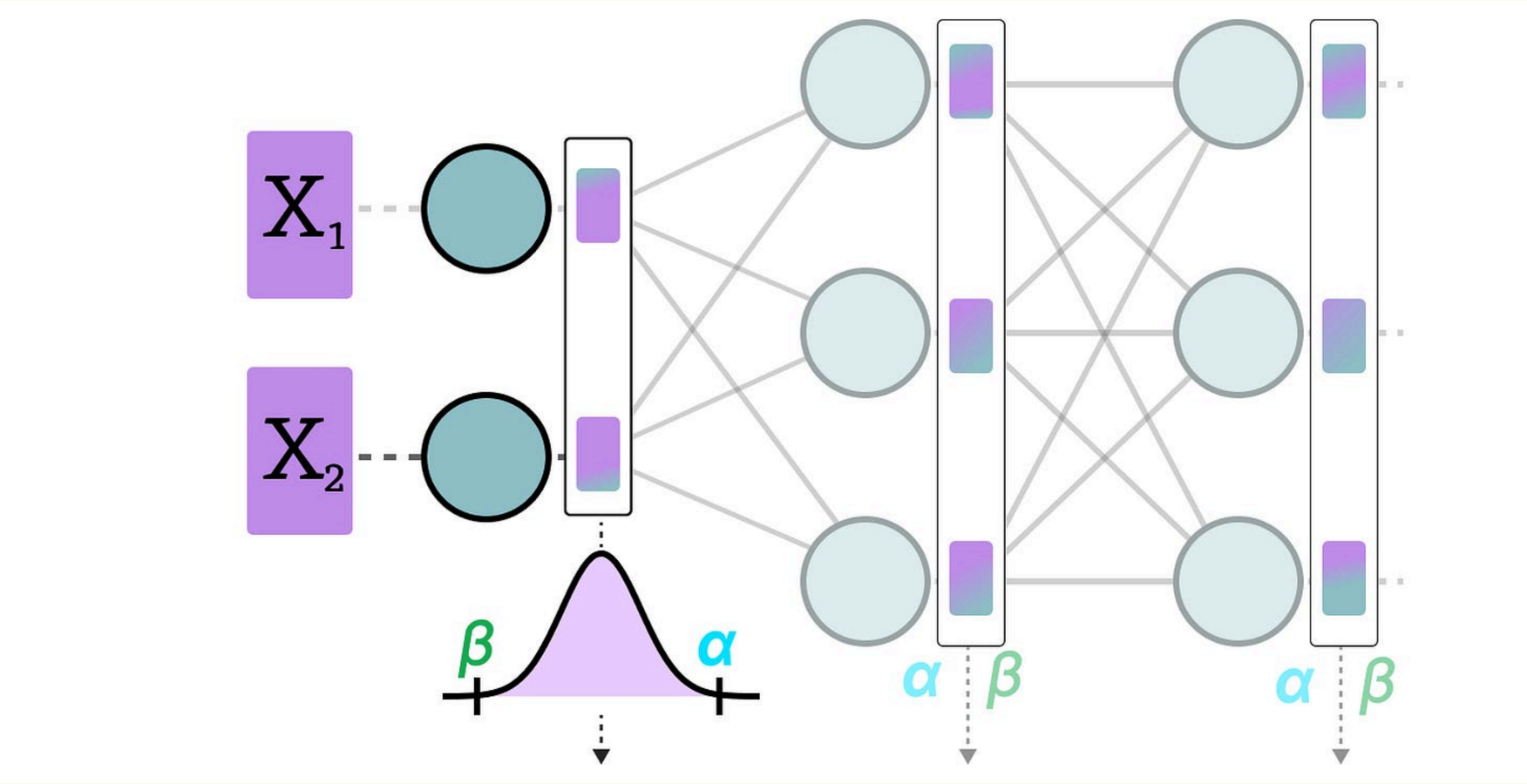
Using the tensor's literal min/max as $[\beta, \alpha]$ is simplest, but a single outlier (as just seen) can force a huge α and waste precision on everything else. An alternative: pick α from a **percentile** of the distribution instead of its true maximum, deliberately clipping rare extreme values:



With an outlier of 999 in the tensor, the full min/max range forces high error onto every normal value; a percentile-based range accepts a large error on the outlier itself in exchange for much lower error everywhere else — the same trade as the clipping example, chosen automatically from the data's distribution rather than a hand-picked cutoff. (Other calibration criteria — minimizing mean-squared error, or KL-divergence between original and quantized distributions — make the same trade using a different statistical target.)

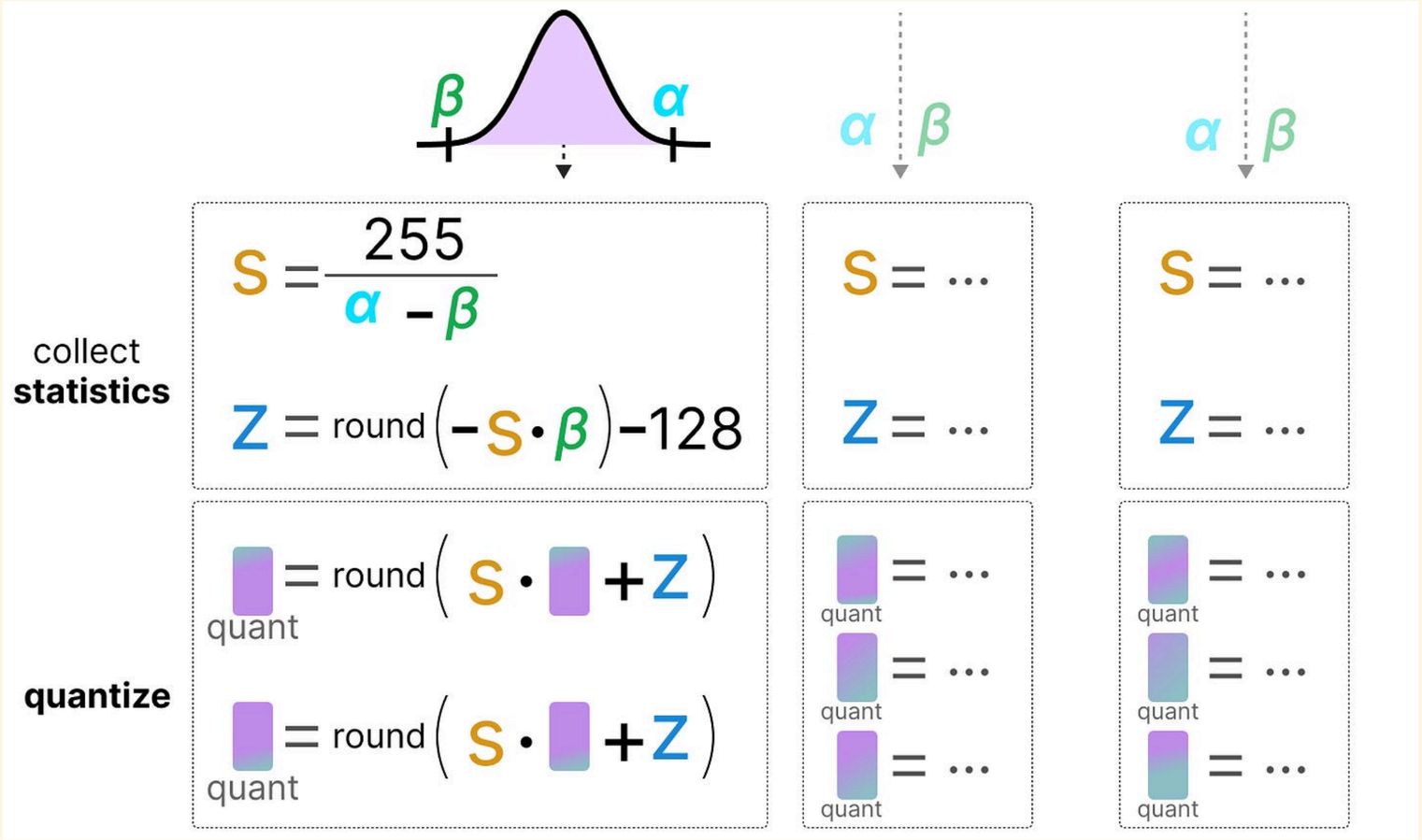
Dynamic quantization

Weights are quantized once, offline, using their fixed range. Activations don't have a fixed range — **dynamic quantization** handles this by computing α, β freshly from the actual values flowing through, on every single forward pass:



No calibration data needed ahead of time, and the range always matches the real input — at the cost of computing s and z live, for every layer, on every inference call.

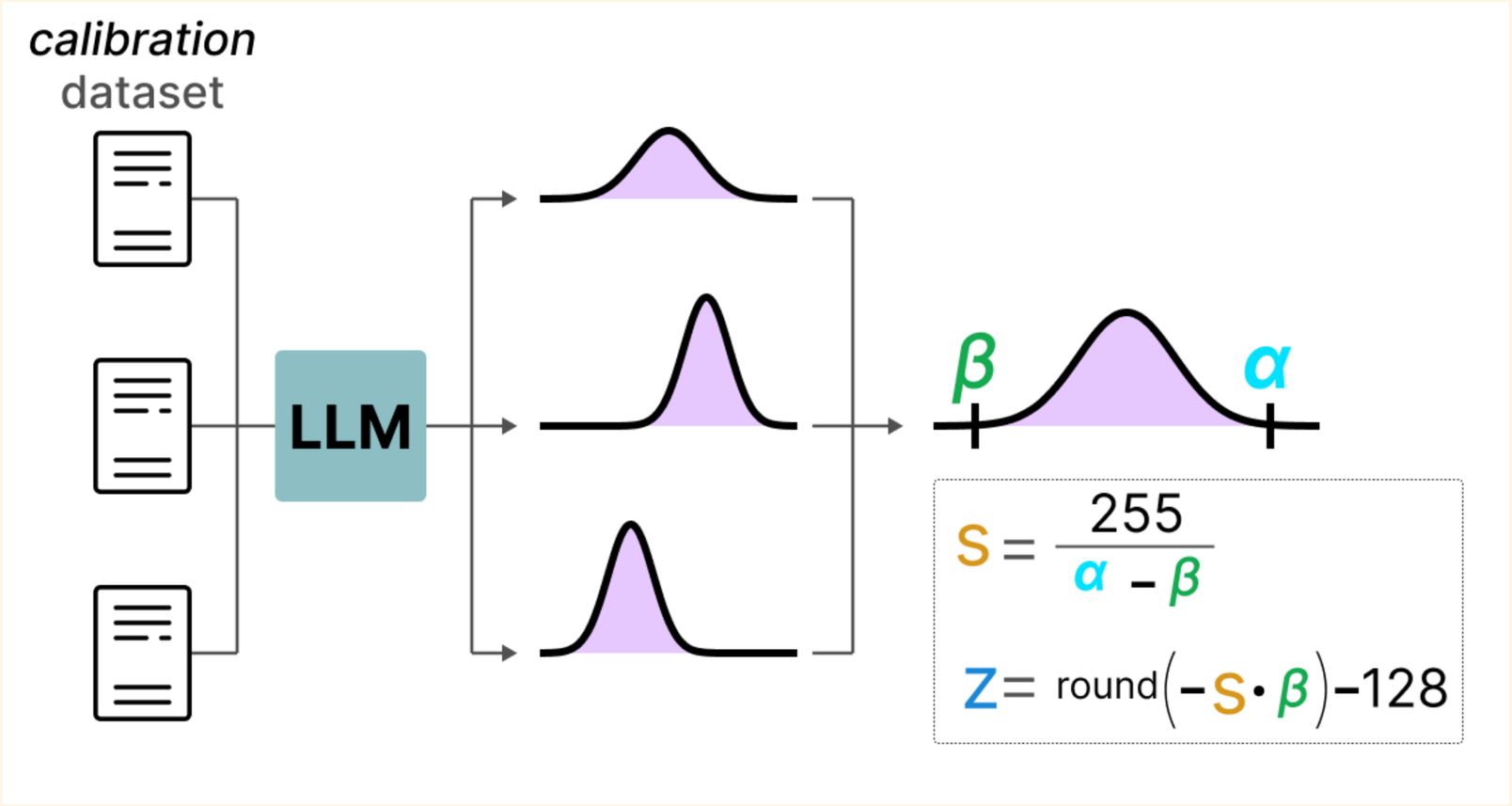
Dynamic quantization: what happens per layer



Each layer runs the same two steps independently, in sequence: collect this layer's α, β from the activations just produced, then quantize them with the resulting s, z — the exact asymmetric formulas from Part 2, just re-run at every layer boundary instead of once in advance.

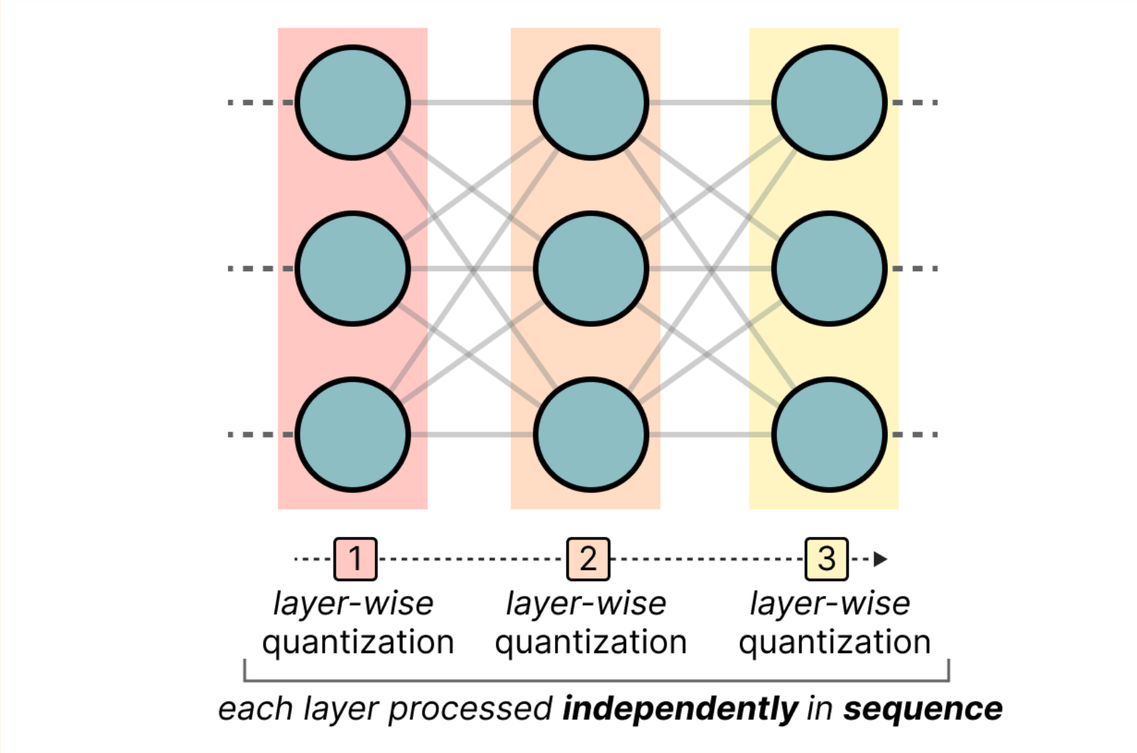
Static quantization

Static quantization avoids computing s, z at inference time at all: run a small calibration dataset through the model once, offline, record the activation distributions it produces, and fix α, β (and hence s, z) from that — before any real inference happens:



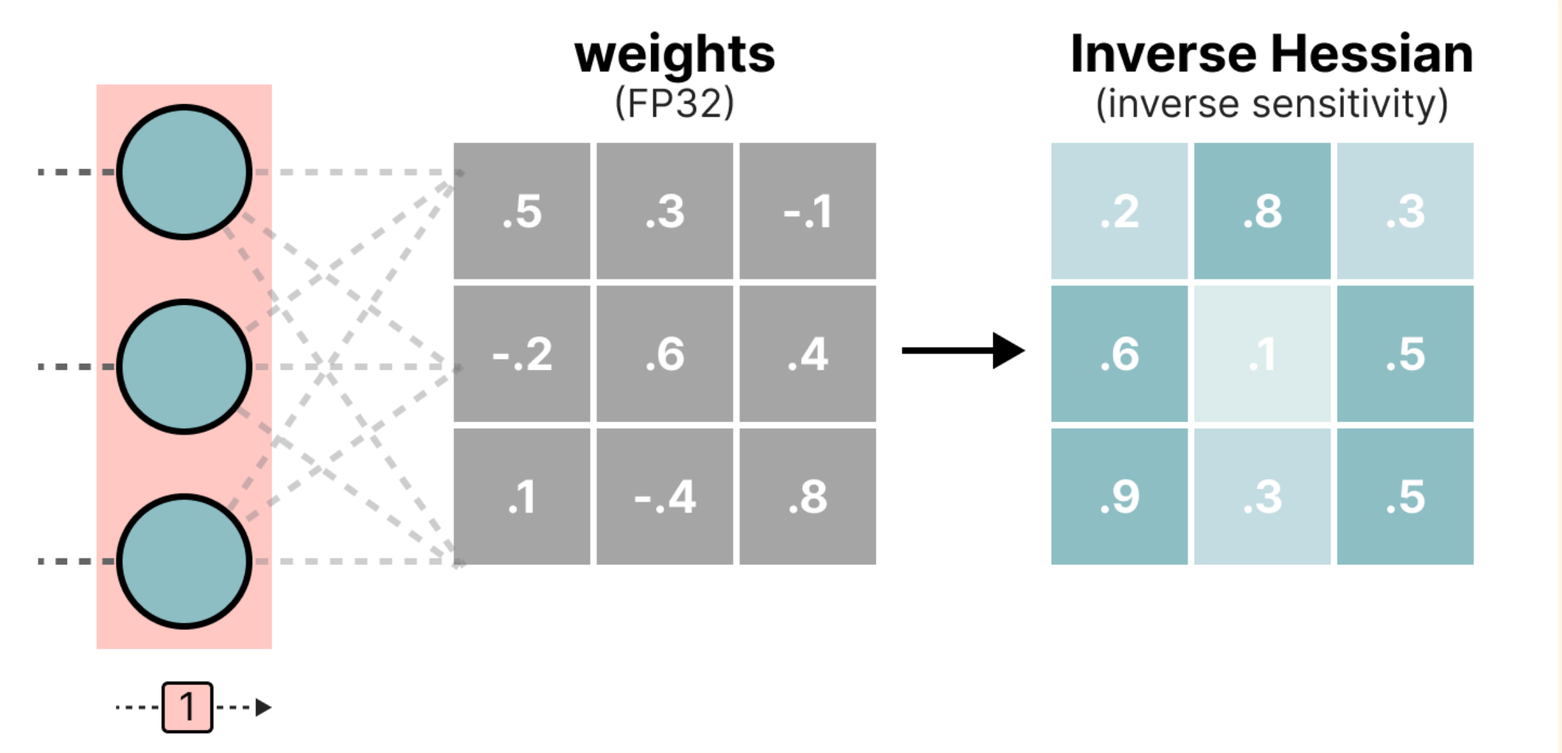
Inference is now as cheap as weight quantization — no per-layer statistics collection — but the fixed range is only as good as how representative the calibration dataset was.

GPTQ: quantizing one layer at a time



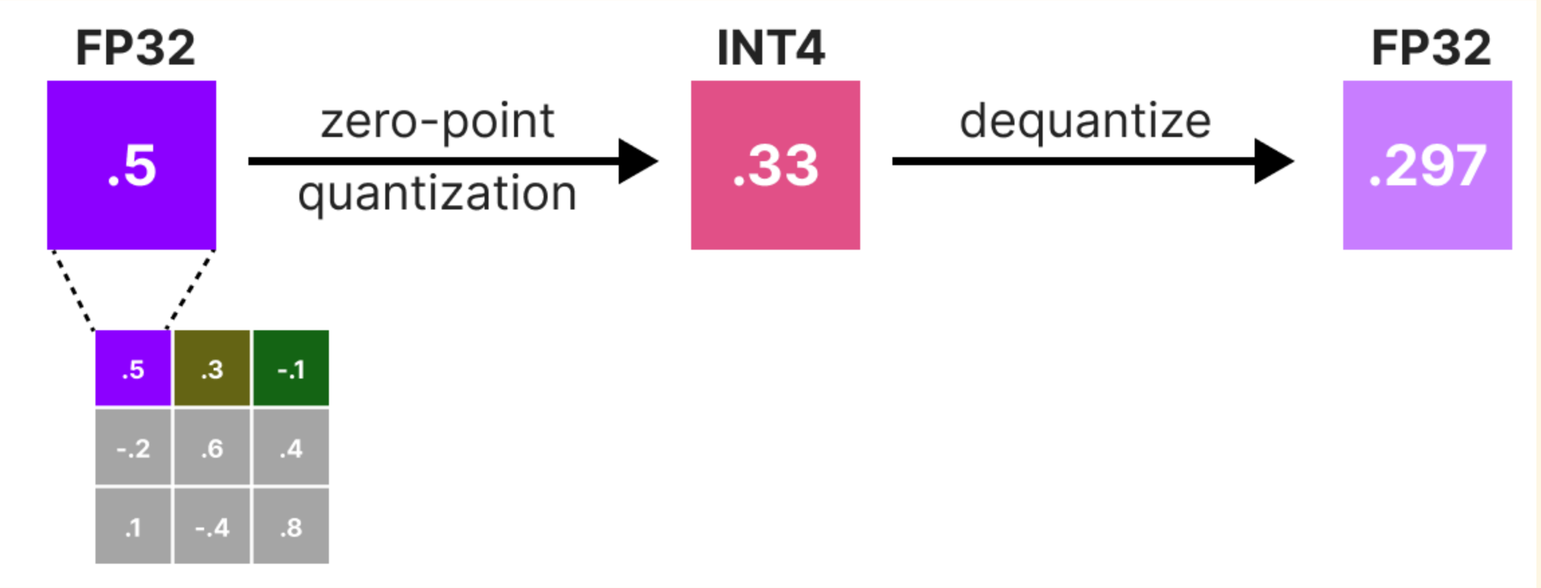
GPTQ quantizes a pretrained model layer by layer, each processed independently and in sequence — turning "quantize the whole network" into a series of smaller, self-contained per-layer problems: *which 4-bit values best reconstruct this one layer's original outputs?*

GPTQ: which weights matter most



Not every weight matters equally to the layer's output — the (inverse) Hessian of the layer's reconstruction error tells GPTQ exactly how sensitive the output is to rounding each particular weight. A weight with a large inverse-Hessian entry can absorb more rounding error for the same output impact; this sensitivity is what later decides how much of one weight's rounding error gets pushed onto the others.

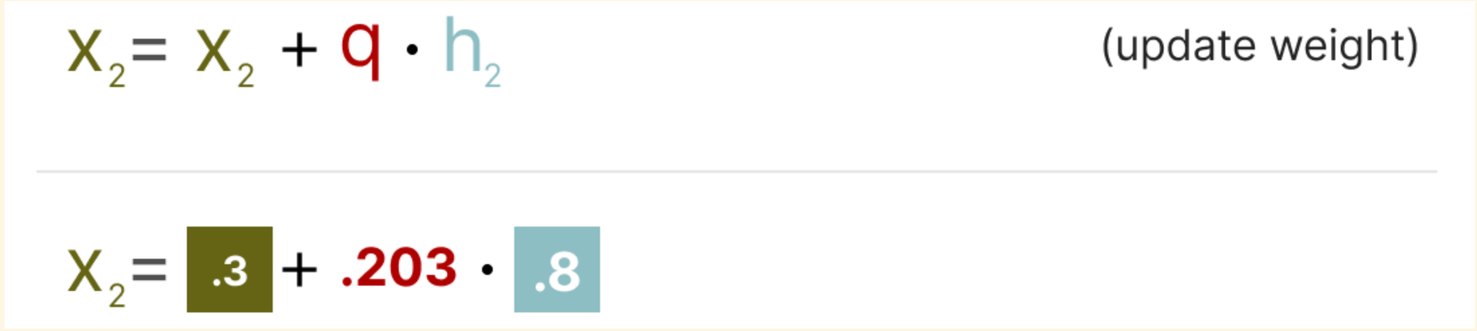
GPTQ: quantize one weight, measure the damage



$$e = w - \hat{w}$$

Quantize the row's first weight ($0.5 \to 0.33$ in 4-bit, dequantized back to $\hat{w} = 0.297$) exactly as in Part 2. $e = w - \hat{w} = 0.5 - 0.297 = 0.203$ is the rounding error this one weight introduced — GPTQ's key idea is that this error doesn't have to be wasted: it can be *redistributed* onto the weights not yet quantized.

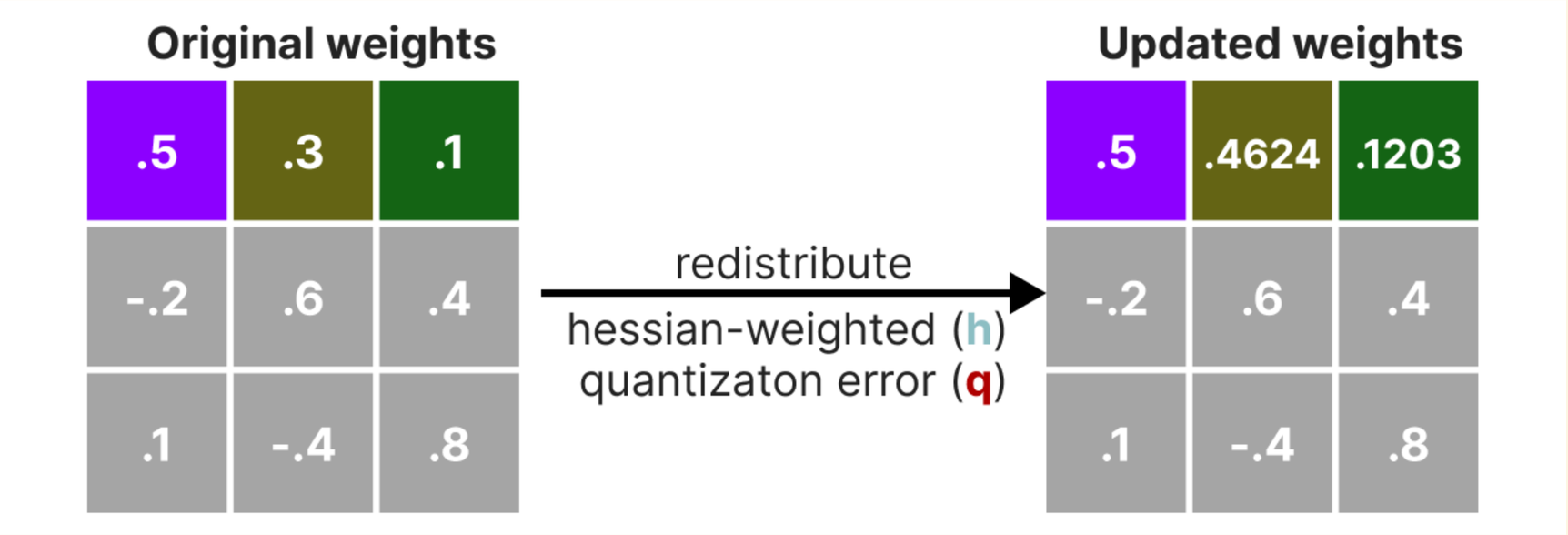
GPTQ: redistributing the error



$$w_j \leftarrow w_j + e \cdot h_j$$

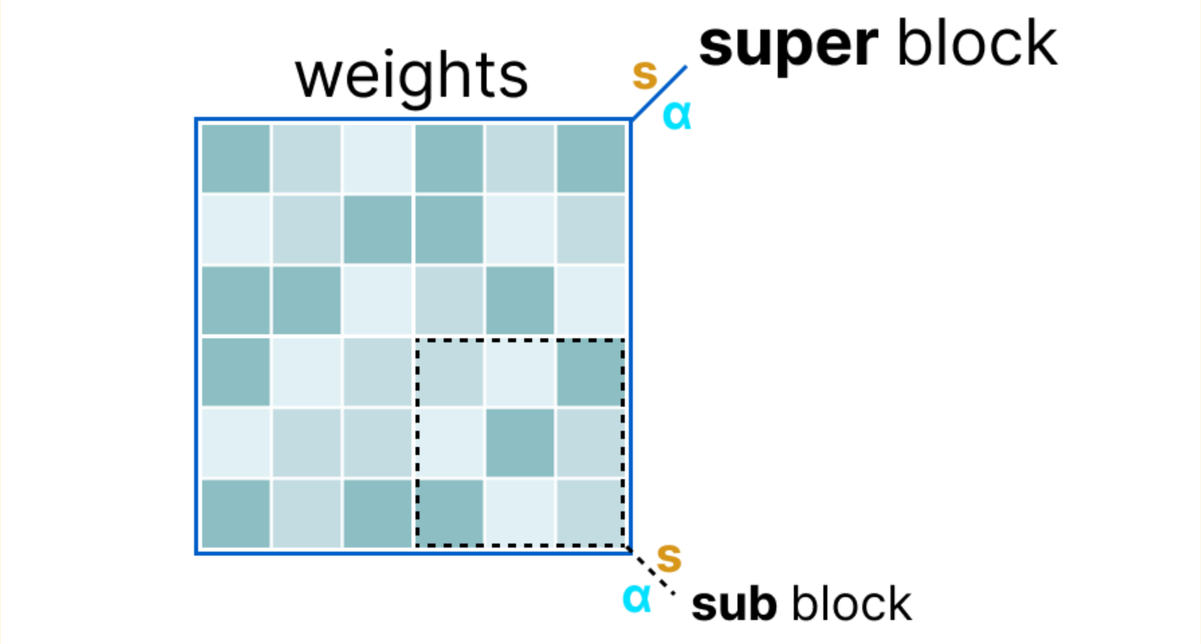
$e = 0.203$ — the rounding error from the previous slide; h_j — how sensitive the output is to weight j (from the inverse-Hessian row). Weight 2, with $h_2 = 0.8$: $w_2 \leftarrow 0.3 + 0.203 \times 0.8 = 0.4624$ — nudged *before* it gets quantized itself, so its own rounding starts from a value that already compensates for weight 1's mistake.

GPTQ: one weight at a time, down the row



Every not-yet-quantized weight in the row absorbs a share of the error, proportional to its own sensitivity h_j, before its turn to be quantized comes up. Repeating quantize-measure-redistribute across the whole row, then the whole layer, is what lets GPTQ push weights to 3–4 bits with far less accuracy loss than quantizing each one independently.

GGUF: blocks inside blocks



Rather than one scale per fixed-size block (Part 2), GGUF's k-quants nest two levels: a **super-block** of weights is split into several **sub-blocks**, and every sub-block gets its own scale — finer-grained than one scale for the whole super-block, without paying for a full scale per individual weight.

GGUF: even the scales get quantized

$$X_{\text{quantized}} = S \cdot X \qquad \text{(absmax quantization)}$$

$$X_{\text{quantized}} = \underbrace{S_{\text{sub}}}_{\substack{\textit{quantized using} \\ S_{\text{super}}}} \cdot X \qquad \text{(absmax quantization)}$$

Ordinary absmax quantization (Part 2) needs one real-valued scale per block. GGUF goes one step further: each sub-block's scale s_{sub} is *itself* quantized, using the super-block's own scale s_{super} — the same double-quantization idea QLoRA uses for its block constants, applied to an integer format instead of NF4.

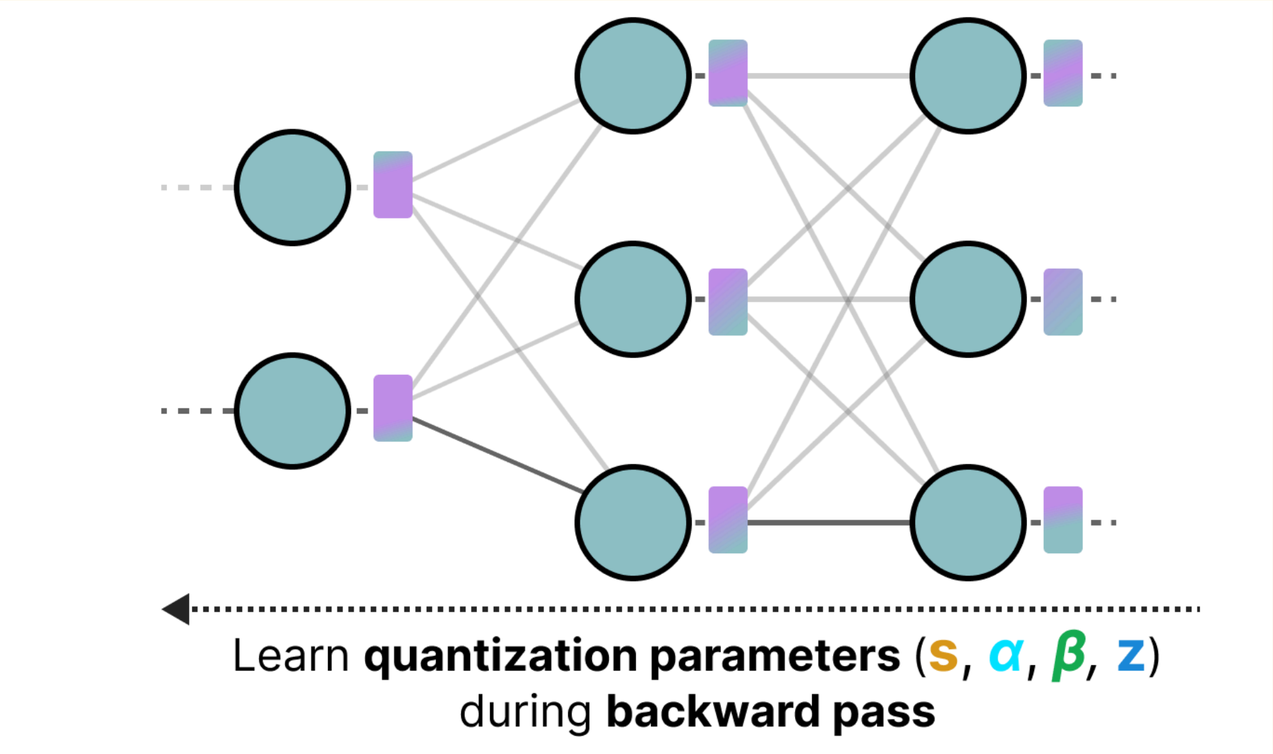
GGUF: quantization levels in practice

Name	Weight quant	Scale (s) quant (“super”)	Scale (s) quant (“sub”)	Bits per weight (w)	# Sub blocks	Weights per block
Q2_K	2 bits	4 bits	2 bits	2.5625	16	16
Q4_K	4 bits	6 bits	4 bits	4.5	8	32
Q6_K	6 bits	8 bits	6 bits	6.5625	16	16

Each named GGUF level trades off weight bit-width against how finely the scales themselves are stored — `Q4_K`'s 4-bit weights end up costing **4.5 bits/weight** once its super- and sub-block scales are counted in, not exactly 4; `Q2_K` and `Q6_K` follow the same accounting at their own bit-widths.

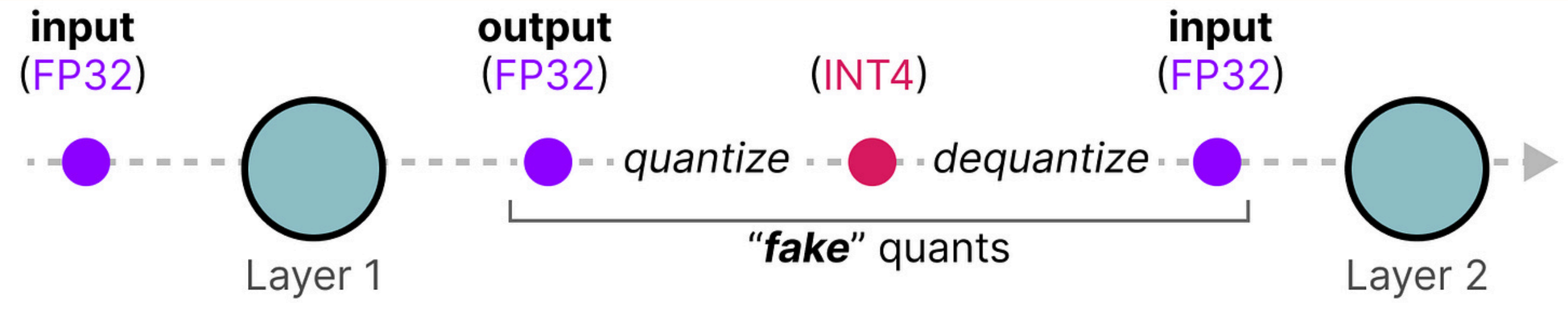
Quantization-aware training

Everything so far quantizes a model **after** it's already trained (post-training quantization, PTQ). **Quantization-aware training** (QAT) instead trains the model already knowing it will be quantized — quantization parameters (s, α, β, z) are learned during the backward pass itself, alongside the weights:



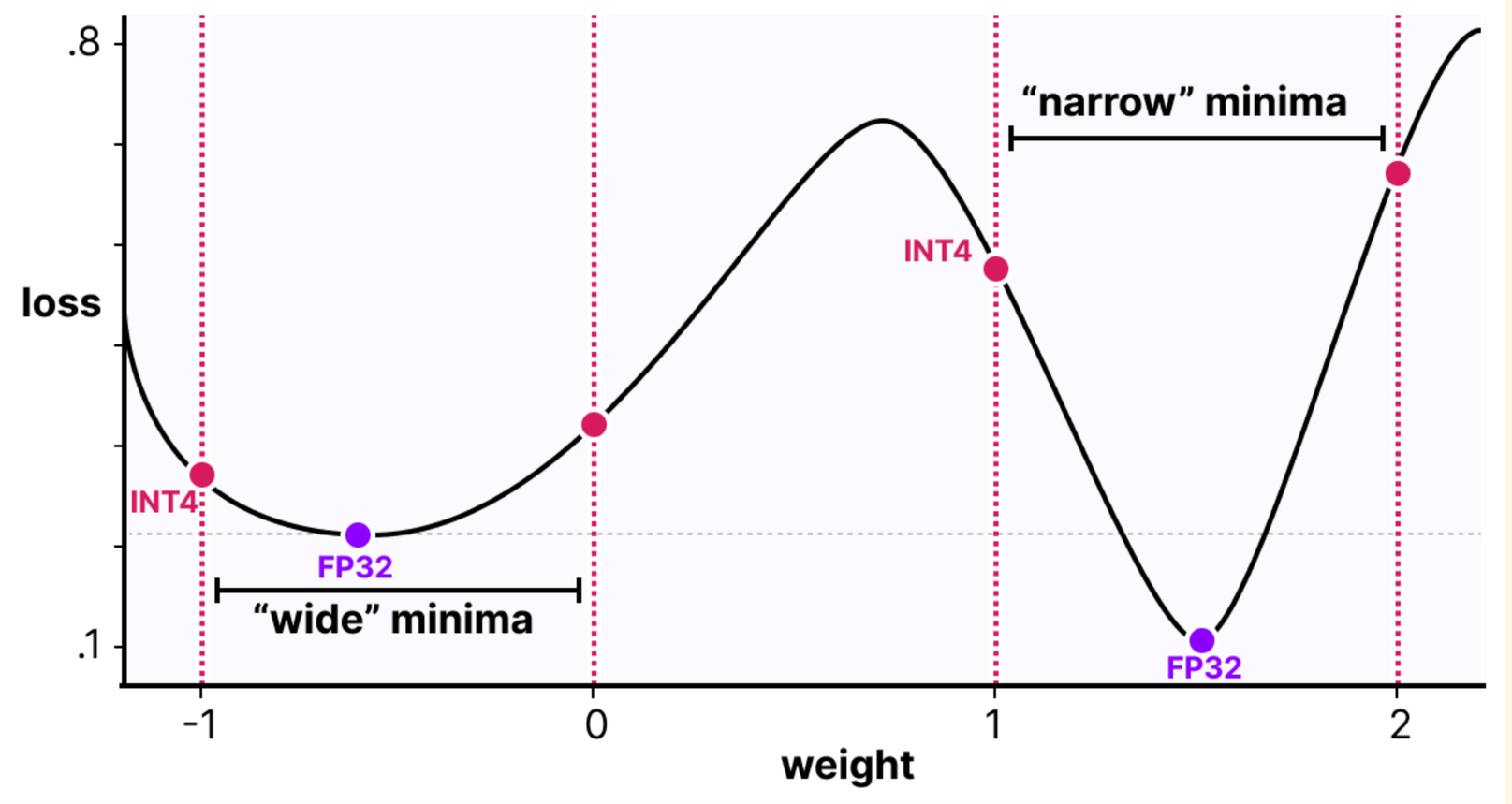
The network's weights adapt *around* the quantization grid, instead of the grid being imposed on weights that never knew it was coming.

Training through a "fake" quantization step



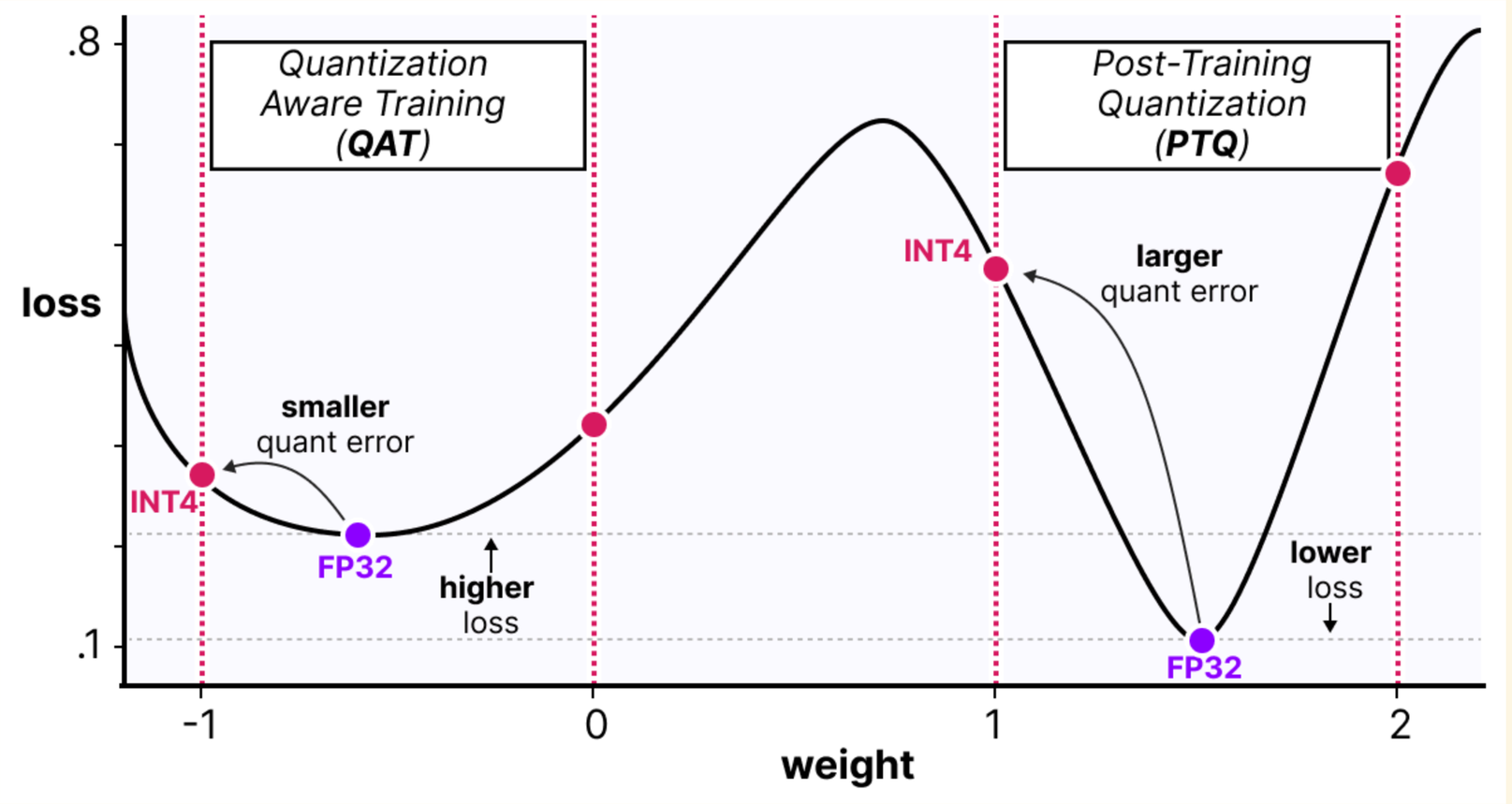
Between layers, every activation is quantized to INT4 and immediately dequantized back to FP32 before continuing — a **"fake" quantization** step. The forward pass sees the same rounding it would see at real low-bit inference; gradients still flow through it in FP32, so the network learns weights that are robust to that specific rounding.

Why this produces better minima



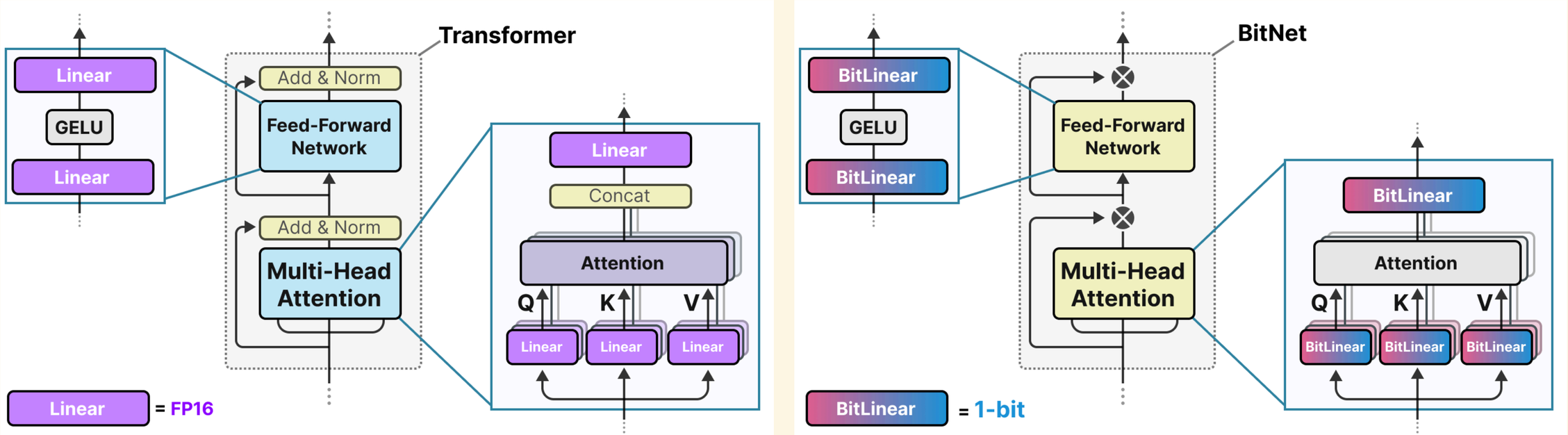
A **wide** minimum stays low-loss even if the weight shifts a bit (as quantization forces it to); a **narrow** minimum's loss shoots up under the same shift. Ordinary training has no reason to prefer one over the other — it just finds the lowest point it can, wide or narrow.

QAT seeks wide minima on purpose



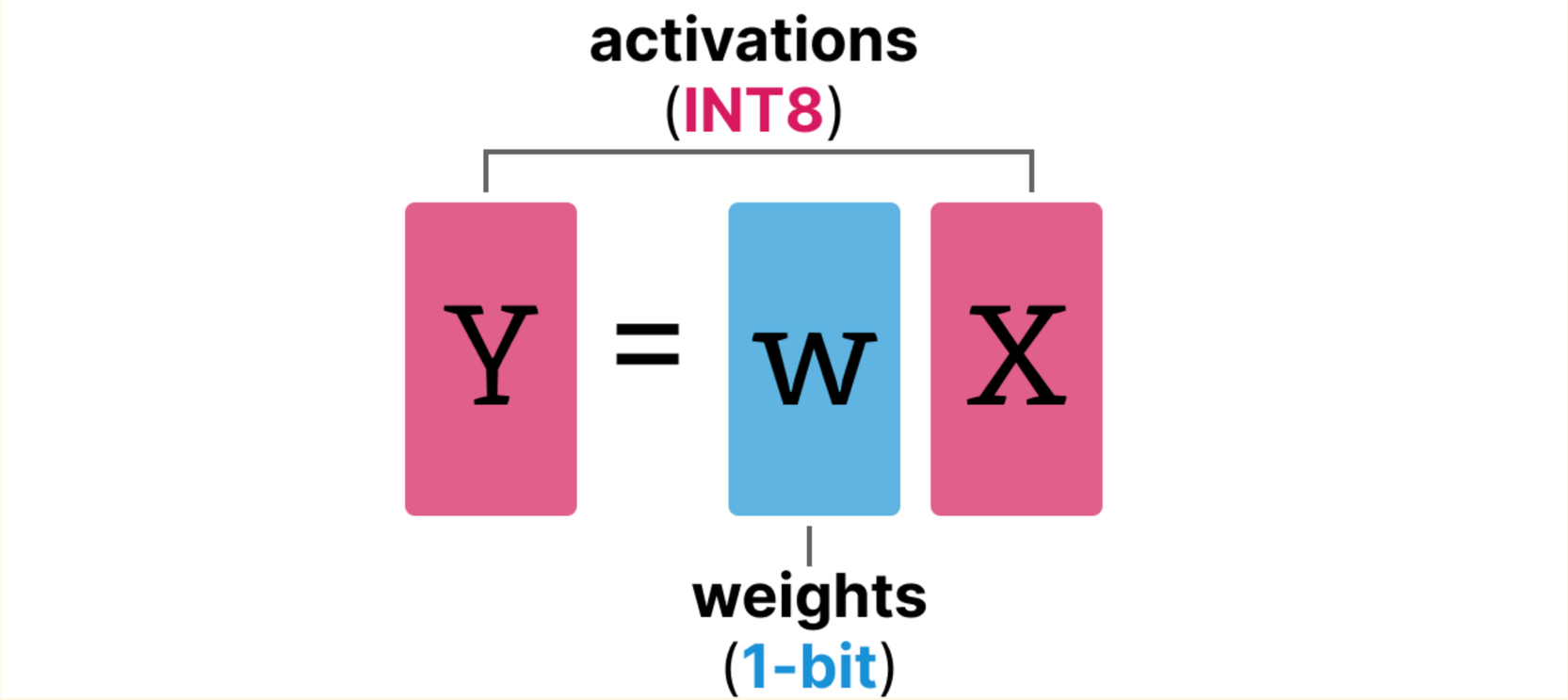
Because QAT's forward pass already includes the rounding step, gradient descent is steered *away* from narrow minima that quantization would wreck, and *into* wide ones that tolerate it — a smaller quantization error for the same bit-width, entirely because training saw the rounding coming. PTQ, quantizing after the fact, has no such guarantee and can land its rounded weight in a narrow, high-loss minimum.

BitNet: replacing Linear with BitLinear



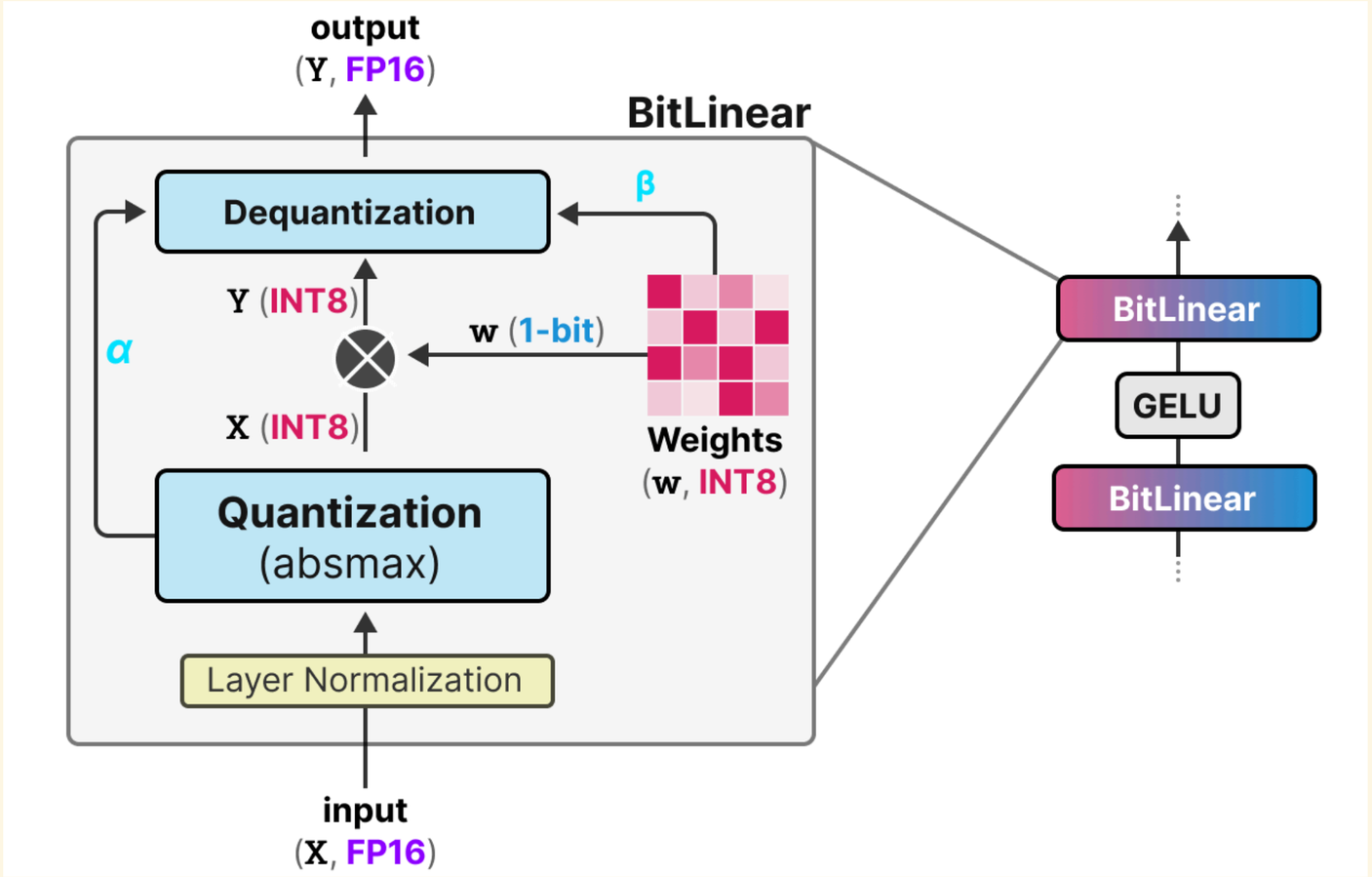
BitNet is QAT taken to its extreme: every ordinary `Linear` layer (FP16 weights) in a Transformer is swapped for a **BitLinear** layer, whose weights are trained to live in just **1 bit** from the start.

BitLinear: 1-bit weights, INT8 activations



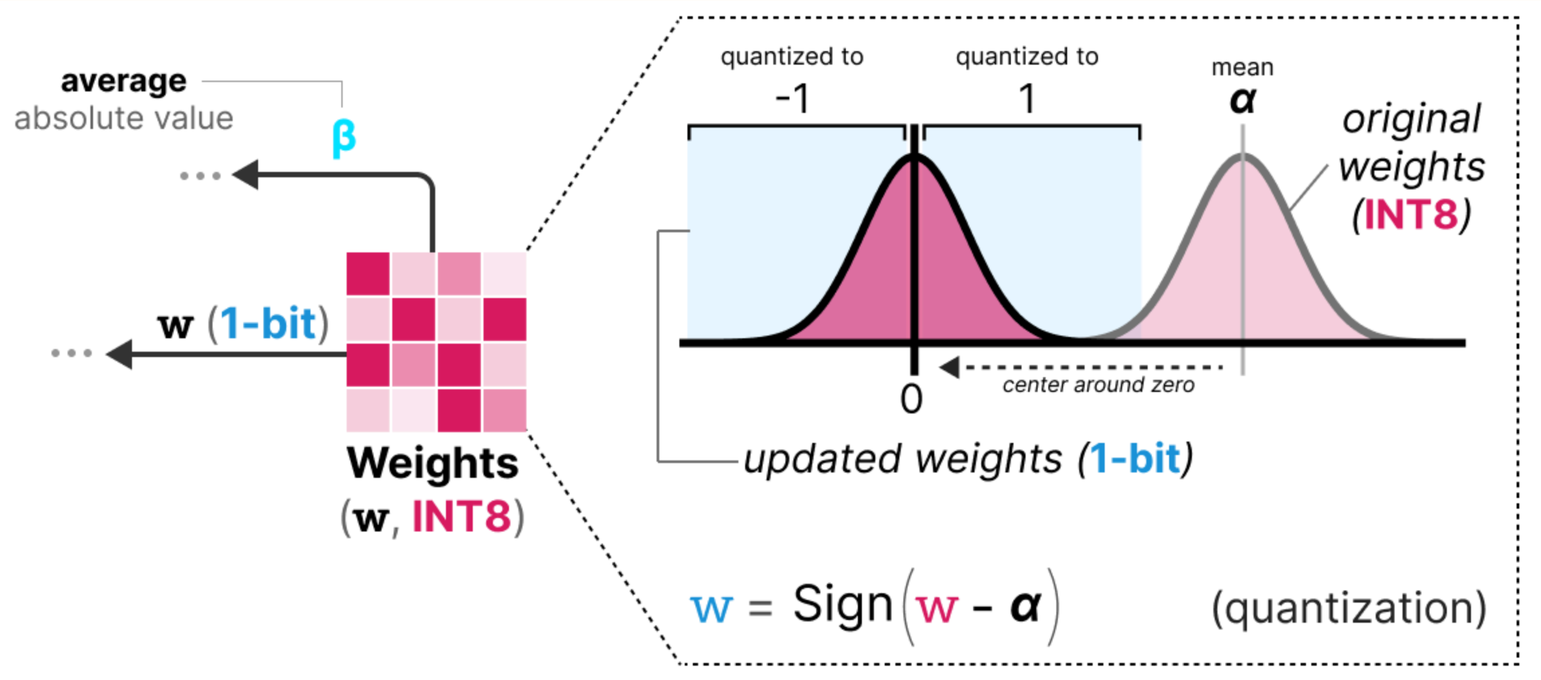
The matrix multiply itself is unchanged, $Y = wX$ — only the precision of w and X changes. Weights are quantized to 1 bit; activations to INT8 (finer than the weights, since activations vary per input and carry more information to preserve).

BitLinear: the full pipeline



Input arrives in FP16, gets layer-normalized, then quantized (absmax, to INT8); weights are held in 1-bit; their product Y (INT8) is dequantized back to FP16 using the tracked scale factors α, β before leaving the layer — every other layer downstream still sees ordinary FP16 activations.

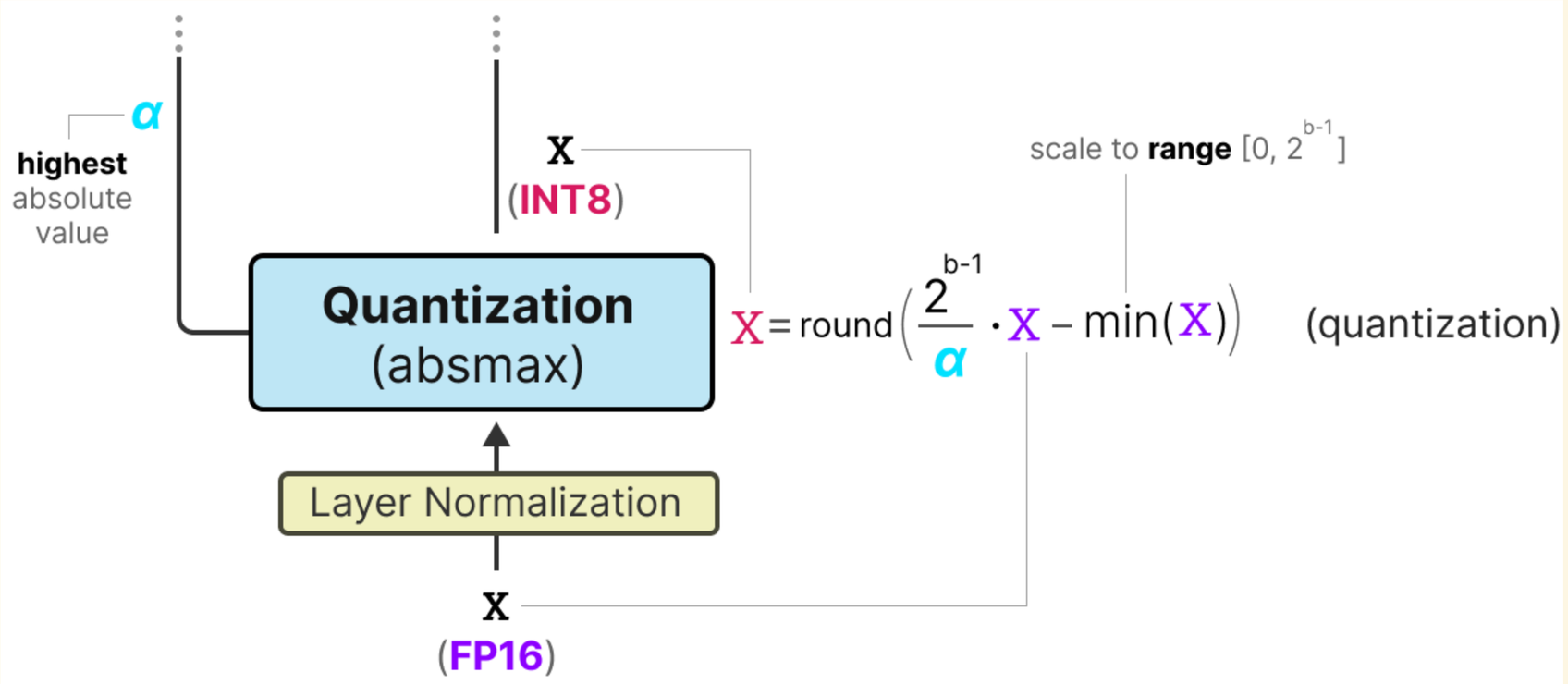
BitNet: quantizing weights to ±1



$$w_{\text{quantized}} = \text{Sign}(w - \alpha)$$

α — the mean weight value in the tensor. Centering on the mean first (not the raw weights) ensures roughly half the weights land on each side of 0 before the sign function collapses everything to exactly -1 or 1 — the full 1-bit weight.

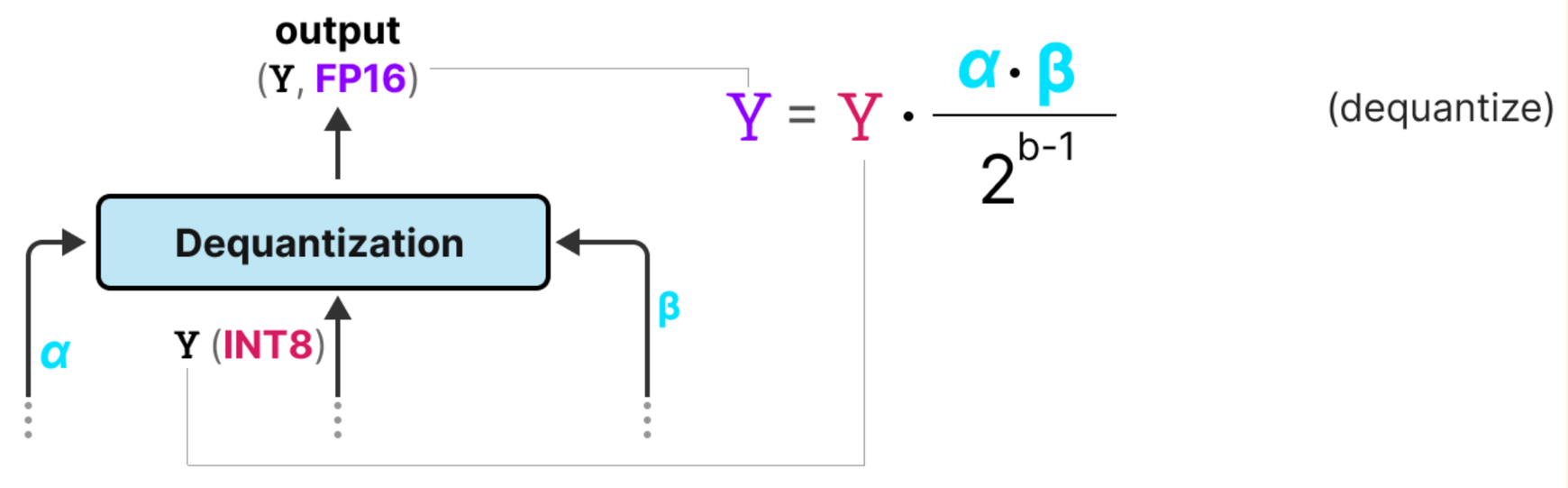
BitNet: quantizing activations to INT8



$$X_{\text{quantized}} = \text{round}\left(\frac{2^{b-1}}{\alpha} \cdot X - \min(X)\right)$$

α — the highest absolute activation value; $b = 8$. Layer normalization runs first, so activations already have a well-behaved, predictable range before this absmax-style scaling maps them into INT8.

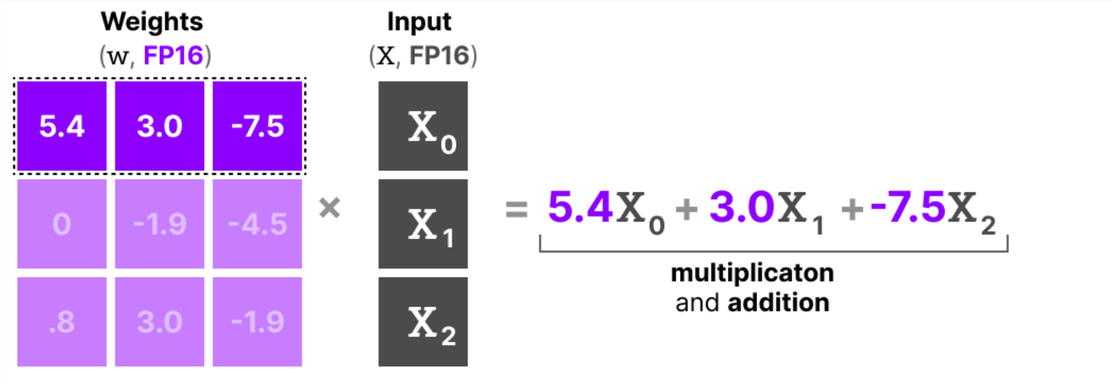
BitNet: recovering an FP16 output

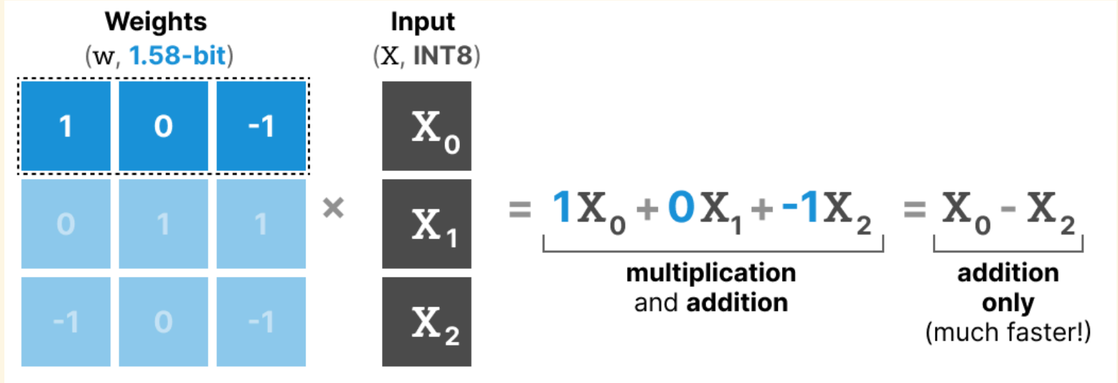


$$Y_{\text{dequantized}} = Y \cdot \frac{\alpha \cdot \beta}{2^{b-1}}$$

α — the weight-quantization scale from a few slides back; β — the activation-quantization scale just used. Multiplying by both undoes both roundings at once, handing the next (ordinary, FP16) layer a normal-looking activation.

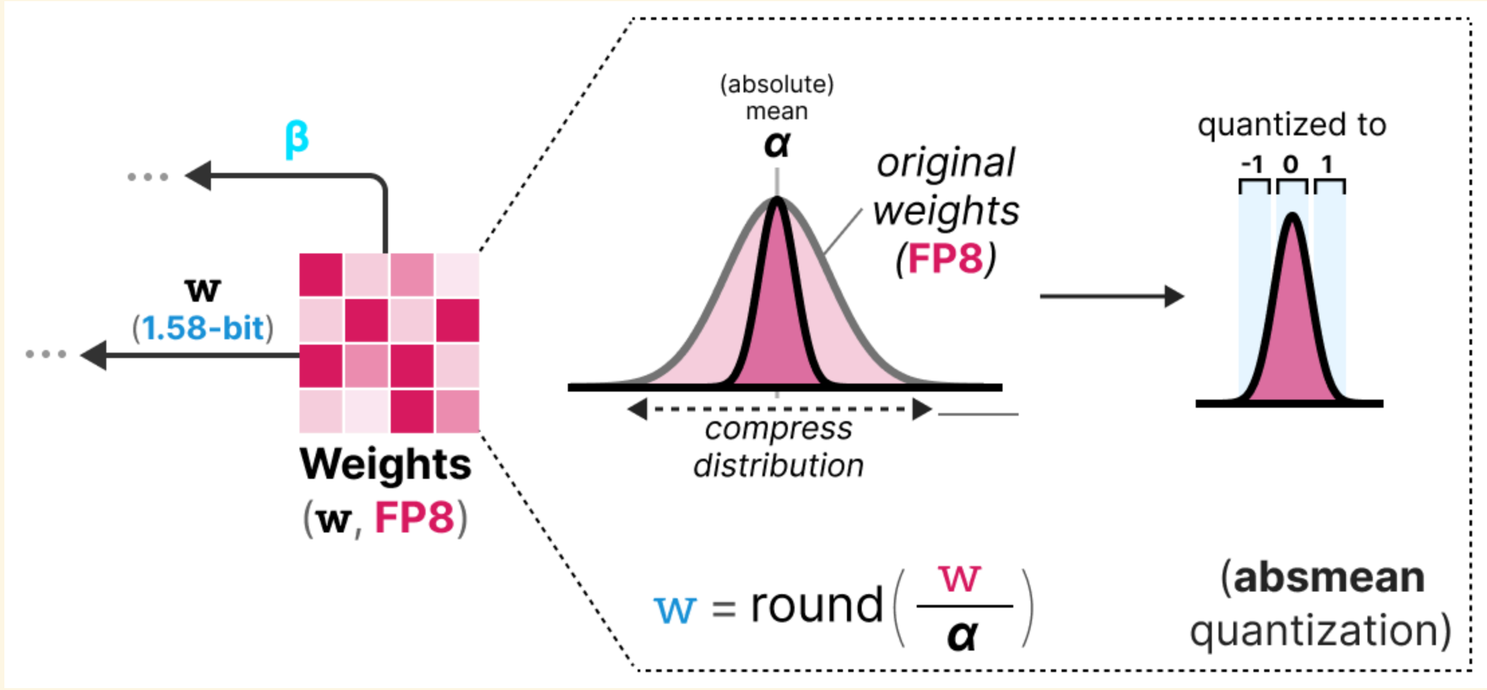
BitNet b1.58: matmul without multiplying





With weights restricted to $\{-1, 0, 1\}$ instead of just $\{-1, 1\}$, each term in the dot product is either $+X_i$, $-X_i$, or dropped entirely — the multiplication disappears, leaving only additions and subtractions. Explicitly allowing 0 is what earns the extra 0.58 bit ($\log_2 3 \approx 1.58$) over BitNet's strict 1-bit version.

BitNet b1.58: absmean quantization



$$w_{\text{quantized}} = \text{round}\left(\frac{w}{\alpha}\right), \qquad \alpha = \text{mean}(|w|)$$

Dividing by the *mean absolute* weight (not the max, as in earlier absmax schemes) compresses the distribution so that most values round to 0, and only the largest round to ± 1 — turning the weight tensor sparse as a side effect of quantizing it.

Reading list

- Grootendorst, *A Visual Guide to Quantization* — the source for every figure in this deck.
- Frantar et al. 2022, *GPTQ: Accurate Post-Training Quantization for Generative Pre-trained Transformers* — arXiv:2210.17323.
- Dettmers et al. 2022, *LLM.int8(): 8-bit Matrix Multiplication for Transformers at Scale* — arXiv:2208.07339.
- Wang et al. 2023, *BitNet: Scaling 1-bit Transformers for Large Language Models* — arXiv:2310.11453.
- Ma et al. 2024, *The Era of 1-bit LLMs: All Large Language Models are in 1.58 Bits* — arXiv:2402.17764.
- Gerganov et al., *GGUF / llama.cpp* — the k-quant format (`Q2_K`, `Q4_K`, `Q6_K`, …) used throughout Part 3.
- NVIDIA, *Model Quantization: Concepts, Methods, and Why It Matters* — a second, industry perspective on the same ideas.